

EEE 306 (January 2026)

Power System I Laboratory

Final Project Report

Section: A1 Group: 03

Feasibility of HVDC Transmission for Integrating a Remote 150 MW Generation Source into the IEEE 30-Bus System

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1 Abstract

Power plants are often built far away from the cities and industries that use the electricity, so the power has to be carried over a long distance before it reaches the main grid. Over such a distance an ordinary high voltage alternating current (HVAC) line runs into trouble. Its resistance wastes real power, its capacitance produces a large amount of unwanted reactive power that pushes the far end voltage above the rated value, and its reactance limits how much power can be sent before the two ends fall out of synchronization. A high voltage direct current (HVDC) link avoids all three of these problems, but it needs a converter station at each end, and those converters inject harmonics into the alternating current network.

This project studies whether an HVDC link is a suitable way to bring the output of a remote 150 MW generating station into the grid over a distance of 750 km. The IEEE 30-bus test system is used as the base network. It is built in ETAP and made stable, and a new bus carrying the remote plant and a local load is added and joined to the main system through a 750 km corridor. The corridor is modelled first as an HVAC line and then as a bipolar HVDC link, and the two options are compared only after each one has been properly compensated, so that the comparison is fair. Three kinds of study are carried out on the model: a load flow study for the voltage profile and the transmission losses, a harmonic study to see what the converters do to the power quality of the network and a contingency study for the sudden loss of the link. Wherever a problem is found, equipment is designed to correct it: shunt reactors and static VAR compensators for the voltage problems, single-tuned passive filters for the harmonics. The report also covers the safety, environmental and social effects of the design, together with a cost estimate prepared for Bangladeshi conditions.

2 Introduction

Electricity is not always produced where it is used. Large hydro plants, big solar parks and big wind farms are usually built in remote places, far away from the cities and industries that need the power. The power then has to travel a long distance before it reaches the main grid. In this project we take exactly this situation: a 150 MW generating station that sits 750 km away from the nearest strong bus of the grid.

If this power is carried by a normal HVAC line, three problems appear at such a long distance. First, the resistance of the line wastes a large amount of real power as heat. Second, and more seriously, the capacitance of the line produces a very large amount of reactive power when the line is lightly loaded. This is the Ferranti effect, and it pushes the voltage at the far end far above the rated value. Third, the reactance of the line limits how much power can be sent before the two ends fall out of step, so the line cannot be loaded up to its thermal limit.

An HVDC link avoids all three problems in a different way. A DC line has no reactive charging current, so there is no Ferranti effect and no stability limit set by line length. The two AC systems joined by the link do not even have to stay in step with each other. The price for this is a converter station at each end, which costs money and, more importantly for this study, injects harmonic currents into the AC network. Whether HVDC is actually worth using depends on the distance: below a certain break-even distance the converter stations cost more than the AC line saves, and above it HVDC wins.

The purpose of this project is therefore to model the same 750 km corridor twice, once as HVAC and once as HVDC, in the same IEEE 30-bus test system, and to compare them fairly. Fairly means that both options are compensated properly before the numbers are compared, because an uncompensated HVAC line at 750 km simply collapses and comparing against it would not be honest.

This problem is a Complex Engineering Problem for the following reasons.

- **Depth of knowledge required (P1).** The work needs power flow theory, long line theory, reactive power control, converter theory, harmonic theory and protection ideas together. No single course topic is enough.
- **Conflicting requirements (P2).** Reducing loss, keeping every bus voltage inside limits, keeping harmonics under the IEEE 519 limits and keeping the cost low all pull against each other. A capacitor bank that fixes voltage can make harmonics worse, and a filter that fixes harmonics also injects reactive power.
- **No obvious solution (P3).** There is no formula that gives the size and place of the shunt reactors, the SVCs and the filters directly. They had to be found by studying the alert reports of ETAP and trying values.
- **Involvement of infrequently encountered issues (P4).** Ferranti effect at 750 km, converter harmonics of order 11 and 13, and the loss of a whole HVDC link as a contingency are not everyday textbook problems.
- **Diverse groups of stakeholders (P5).** The utility, the plant owner, the people living along the 750 km corridor and the regulator all have different interests.
- **Interdependence (P6).** The choice of link technology changes the voltage profile, which changes the reactive support, which changes the harmonic response, which changes the contingency behavior. Nothing can be designed alone.

Four possible ways of solving the problem were considered.

- HVAC line with shunt compensation. Keep the AC line and add shunt reactors and SVCs to cancel the charging reactive power. This is cheap and simple but the compensation equipment is large (about 420 MVar in our case) and the loss stays higher.
- Line-commutated converter (LCC) HVDC. A thyristor based bipolar link. It has low loss, high power rating and low cost per km, but it needs a strong AC system at both ends, it absorbs reactive power and it produces strong 11th and 13th order harmonics. This is the option modelled in this project.
- Voltage source converter (VSC) HVDC. An IGBT based link that can control real and reactive power independently and works with weak AC systems, but it costs more and has higher converter losses.
- Series compensation of the AC line. Series capacitors reduce the effective reactance, but they do not solve the charging current problem and they bring the risk of sub-synchronous resonance.

Of these four, the first two were built and studied in ETAP, because the aim of the project is a fair comparison between a compensated AC corridor and a DC corridor at the same distance. The VSC link was set aside because its higher cost and higher converter loss make it hard to justify for a simple point-to-point overhead corridor, and it is taken up again as future work in Section 9. Series compensation was set aside because it reduces the effective reactance but does nothing about the line charging current.

3 Design

3.1 Problem Formulation (PO(b))

3.1.1 Identification of Scope

The following work is inside the scope of this project.

- Building the standard IEEE 30-bus system in ETAP and making it stable, so that it can be

used as a realistic base network.

- Adding a new Bus 31 with a 150 MW generator and a 20 MW (21 MVA) local load, and joining it to Bus 1 of the IEEE 30-bus system through a 750 km corridor.
- Modelling that corridor first as an HVAC line and then as a 660 kV bipolar HVDC link.
- Designing the reactive compensation (shunt reactors and SVCs) needed to make each option operate inside normal voltage limits.
- Comparing the two options on line loss, total system loss and voltage profile.
- Running a harmonic load flow on the HVDC case and designing passive filters so that the distortion meets IEEE Std 519.
- Running a contingency study for the sudden loss of the HVDC link and analysing the stability.
- Making a rough cost estimate of the compensation and filter equipment.

The following work is outside the scope of this project.

- Transient and electromagnetic transient (EMT) studies of the converters. Only steady state and harmonic load flow studies were done.
- Detailed converter firing control, tap changer control and protection relay settings.
- Detailed tower, conductor, insulator and right-of-way design.
- Full life-cycle cost analysis of the converter stations themselves.

3.1.2 Literature Review

Seven main sources were studied before the simulation work started.

- **Al-Taree and Altahir [1]** ran an optimal power flow on the same IEEE 30-bus system with an HVDC link and compared it with the pure AC grid. They reported lower losses and a better voltage profile with HVDC. This paper is the closest to our work and gave us the idea of using the IEEE 30-bus system as the test bed.
- **Okba et al. [2]** gave a general review of HVDC transmission. From this we took the standard reasons for choosing HVDC: no charging current, no stability limit set by distance, asynchronous interconnection, and the existence of a break-even distance of roughly 600-800 km for overhead lines.
- **Yang et al. [3]** described bipolar HVDC lines and the use of the DC resistance of the line in protection. We used their bipolar arrangement and the idea that the loss of a bipolar DC line is $2I^2R$, not $3I^2R$ as in a three phase AC line.
- **Kimbark [4]** is the classical text on direct current transmission. Chapter 8 gives the harmonic orders produced by a converter and the design of single-tuned filters, which is exactly the method we followed for our filters.
- **Kundur, Balu and Lauby [5]** provided the typical per kilometer line constants (R, X and B) for an EHV overhead line, which we used for the 750 km line, and the theory of the Ferranti effect and shunt reactor compensation.
- **Shelar, Bankar and Bakre [7]** reviewed how IEEE Std 519 has changed from 1981 to 2022. From this we took the present voltage distortion limits, in particular the 1.5% THD limit for buses above 161 kV and the 2.5% limit for 69-161 kV buses, which our filters had to meet.

The gap we found is that most papers compare an HVDC link with an uncompensated AC line, which makes HVDC look much better than it really is. Very few compare a properly compensated AC line with a properly compensated DC link, and fewer still follow the study through to harmonic filter design and contingency analysis on the same network. That gap is what this project tries to fill.

3.1.3 Formulation of Problem

The problem can be stated in one sentence: a 150 MW generator at a new Bus 31 has to deliver its surplus power to Bus 1 of a stable IEEE 30 – bus system over a distance of 750km, and we have to decide whether the corridor should be HVAC or HVDC and design whatever extra equipment each choice needs.

The data given to us are listed below.

- **Base network:** IEEE 30-bus system, 132 kV and 33 kV levels, 50 Hz.
- **New generation at Bus 31:** 150 MW, generator terminal voltage 13.8 kV, stepped up by GSU – 31 (150 MVA) to 132 kV and then by T31 (200 MVA) to 400 kV.
- **Local load at Bus 31:** 20 MW (21.051 MVA), so about 130 MW is available for export.
- **Corridor length:** 750 km, with per kilometer constants $R = 0.036 \Omega/km$, $X = 0.31 \Omega/km$ and $Y = 3.77 \mu S/km$ at 400 kV.
- **HVDC option:** Bipolar link, 660 kV, transmission limit 100 MW, 27Ω resistance per conductor.

The design must satisfy the following constraints.

- Every bus voltage must stay between 0.95 and 1.05 per unit, and no bus may be flagged as critical by ETAP.
- No transformer, generator or line may be overloaded.
- Voltage THD must satisfy **IEEE Std 519**: 1.5% for buses above 161 kV and 2.5% for buses between 69 kV and 161 kV.[15]
- The system must survive the sudden loss of the new link without any critical alarm.
- The extra equipment should be as small and as cheap as possible.

3.1.4 Analysis

Before any simulation was run, a hand analysis was done to find out what to expect at 750km. This analysis decided the whole design.

Charging reactive power of the AC line: The total shunt susceptance of the corridor is $B = 3.77 \mu S/km \times 750km = 2.83 mS$. At 400 kV the reactive power generated by the line capacitance alone is $Q = V^2 \times B = (400kV)^2 \times 2.83mS \approx 452 MVAR$. This is more than three times the real power the line has to carry. This single number explains everything that went wrong in the uncompensated HVAC case: the far end voltage rose to 209.6% of nominal and the transformer was overloaded to 389.2%. It also told us in advance that roughly 400-450 MVAR of shunt reactors would be needed, and the final design used 420 MVAR.

Resistive loss of the AC line: The total resistance is $R = 0.036 \Omega/km \times 750km = 27 \Omega$ per phase. Carrying about 98.6 MW at 400 kV, the current is $I = \frac{P}{\sqrt{3} \times V} \approx 142A$, so the expected loss is $3I^2R \approx 1.64MW$. The compensated ETAP simulation gave $1.543MW$, which is very close and confirms that the model is correct.

Resistive loss of the DC line: A bipolar DC line has only two conductors and no reactive current, so the loss is $2I^2R$ where $I = \frac{P}{V}$. Carrying 100 MW at 600 kV, $I \approx 167 A$ and the expected loss is about 1.5 MW. ETAP reported 1.264 MW for the compensated HVDC case, again the same order of magnitude.

Interpretation of the analysis: The hand calculations indicate that at 750 km the two options should perform similarly, with the direct current alternative holding a modest advantage. This advantage does not arise from the resistive loss, since both corridors have the same total resistance of 27Ω . It arises from the absence of line charging power. The alternating current corridor generates approximately 452 MVar of reactive power that must be absorbed by 420 MVar of shunt reactors, and the associated reactive current flows through every transformer along the path, producing

additional loss in each. The direct current corridor carries no such current. Because 750 km lies **close** to the accepted break-even distance for overhead transmission, only a small margin was expected, and the simulation results support this: the link loss is 1.543 MW for the compensated alternating current option against 1.264 MW for the direct current option, and the corresponding total system losses are 11.072 MW and 10.817 MW.

Cost of HVDC: The benefit is not free. The converters draw reactive power and inject harmonics of order 11 and 13, which had to be filtered, and the link is a single point of failure whose loss had to be studied as a contingency. Both of these are dealt with later in the design.

3.2 Design Method (PO(a))

This section lists the equations that were used to size every piece of equipment in the design.

- (a) **Long transmission line.** The 750 km line is a long line, so it was represented by its total series impedance $Z = (R + jX) \times l = (0.036 + j 0.31) \times 750 \text{ km} = 27 + j232.5 \Omega$ and its total shunt susceptance $B = 3.77 \mu\text{S}/\text{km} \times 750 \text{ km} = 2.83 \text{ mS}$, split equally between the two ends in the nominal π – model. ETAP uses the exact long line model with the correction factors.
- (b) **Ferranti effect and shunt reactor sizing.** The reactive power produced by the line capacitance is $Q_C = V^2 B$. A shunt reactor connected to a bus absorbs $Q_L = \frac{V^2}{X_L}$ so the reactance needed for a given rating is $X_L = \frac{V^2}{Q_L}$. For the 170 MVar reactor at Bus31_400 this gives $X_L = \frac{(400 \text{ kV})^2}{170 \text{ MVar}} \approx 941 \Omega$. The reactors were split between the two ends (170 MVar at the sending end and 250 MVar at the receiving end) so that neither end sees a large voltage rise.
- (c) **AC line loss.** $P_{\text{loss}} = 3I^2 R$, where $I = \frac{S}{\sqrt{3}V_{LL}}$. Because the current has both a real and a reactive part, any reactive power flowing in the line adds to the loss even though it carries no useful energy. This is the main reason the compensation was placed close to the line ends.
- (d) **DC link.** For a bipolar DC line, $P_{dc} = V_{dc} \times I_{dc}$ and the loss is $P_{\text{loss}} = 2I^2 R$, where R is the resistance of one conductor. There is no reactive component, no skin effect worth counting and no charging current, so for the same conductor the DC loss is lower.
- (e) **Converter harmonics.** A 12-pulse line-commutated converter produces characteristic harmonics of order $h = 12k \pm 1$, that is the 11th, 13th, 23rd, 25th and so on. A 6-pulse group would also produce the 5th and 7th. The harmonic load flow in ETAP confirmed that the 11th and 13th were dominant in our network.
- (f) **Total harmonic distortion.** THD is the ratio of the RMS value of all the harmonic components to the fundamental, expressed as a percentage. IEEE Std 519 sets a limit of 1.5% for buses above 161 kV, 2.5% for 69-161 kV, 5.0% for 1-69 kV and 8.0% for buses at or below 1 kV, with individual harmonic limits of 1.0%, 1.5%, 3.0% and 5.0% respectively.[15]
- (g) **Single-tuned filter design.** Each filter is actually a series R-L-C branch. The capacitor is chosen first from the reactive power it has to supply, $X_{C1} \approx \frac{V_{LL}^2}{Q_f}$. The filter is then tuned to the harmonic of n^{th} order by making the inductive and capacitive reactance equal at that frequency, $X_L(\omega_n) = X_C(\omega_n)$, which gives $X_L = \frac{X_C}{n^2}$. The characteristic impedance at resonance is $X_0 = \sqrt{\frac{L}{C}}$, and the damping resistance follows from the Quality – factor, q as $R = \frac{X_0}{q}$. A quality factor of 20 ($q = 20$) was used, which gives a sharp filter.

3.3 Circuit Diagram

The single line diagrams below were drawn in ETAP. Figure 1 is the base IEEE 30-bus system after it was made stable. Figure 2 shows the same system with the new Bus 31, the 150 MW generator, the local load and the 750 km HVAC corridor added. Figure 3 shows the final arrangement in which the corridor between Bus31_400 and BUS1_400 has been replaced by the bipolar HVDC link.

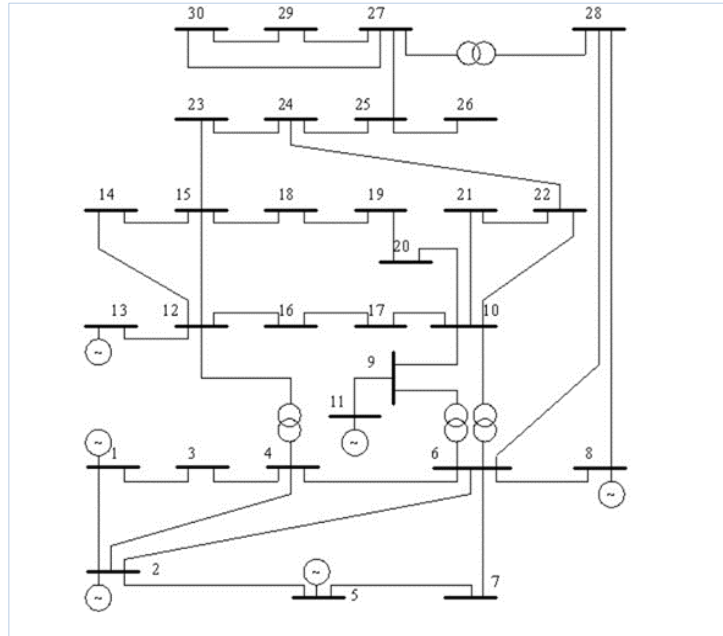


Figure 1: IEEE - 30 Bus System

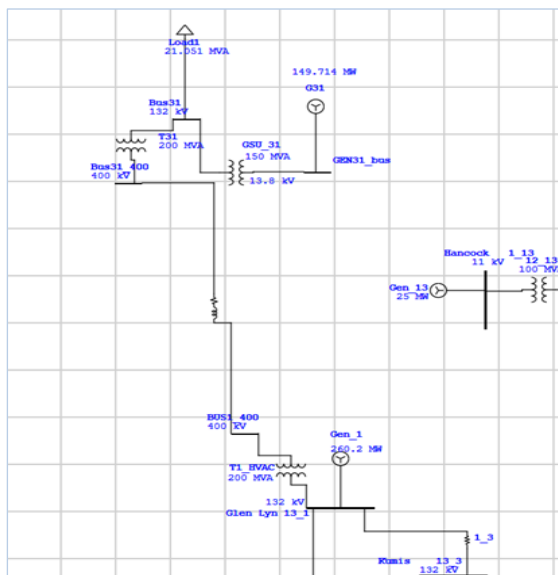


Figure 2: Bus 31 and the 150 MW plant connected to Bus 1 through the 750 km HVAC corridor.

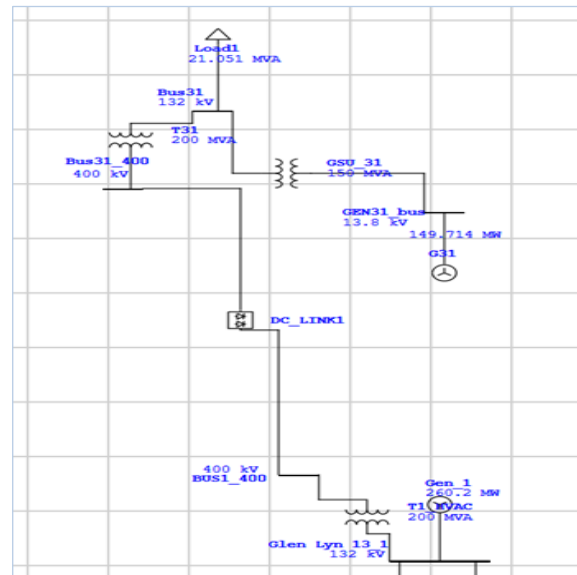


Figure 3: The same corridor rebuilt as a 660 kV bipolar HVDC link (DC_LINK1), with the two harmonic filters HF7 and HF3.

3.4 Simulation Model

All the modelling and all the studies were done in ETAP. The base IEEE 30 – Bus data (bus voltages, line impedances, generator ratings and loads) were entered first and a Newton-Raphson load flow was run until the system was stable with no critical alarm. Five buses (24, 26, 29, 30 and Glen Lyn 13_1) were left marginal, between 96.0% and 103.0% of nominal, and the remaining twenty-five buses were fully inside limits.

The extra equipment was then added on top of this base case. The main data used in the model are given in the tables below.

Table 1: Data of the new generation and the 750 km HVAC corridor.

Item	Value
Line length	750 km
Resistance	0.036 Ω /km (27 Ω total)
Reactance	0.31 Ω /km (232.5 Ω total)
Shunt admittance	3.77 μ S/km (2.83 mS total)
Line voltage	400 kV
Generator G31	150 MW, 13.8 kV
Local load at Bus 31	20 MW (21.051 MVA)
GSU_31 / T31 / T1_HVAC	150 MVA / 200 MVA / 200 MVA

Table 2: Data of the HVDC link.

Item	Value
Configuration	Bipolar
DC voltage	660 kV
Transmission limit	100 MW
Resistance per conductor	27 Ω
Length	750 km
Converter rating (DC_LINK1)	150 / 100 / 150 MVA

Table 3: Reactive compensation designed for each case.

Case	Compensation used
HVAC	170 MVar shunt reactor at Bus31_400
HVAC	250 MVar shunt reactor at Bus1_400
HVAC	30 MVar absorbing SVC at Bus 1
HVDC	45 MVar absorbing SVC at generator Bus 1
HVDC	20 MVar absorbing SVC at Bus31_400
Contingency	SVC 5.89 MVar at Bus 26, 6.18 MVar at Bus 29, 8.84 MVar at Bus 30

Table 4: Parameters of the two single-tuned harmonic filters.

Parameter	Filter HF7 (Bus 31)	Filter HF3 (Bus 30, Glen Lyn 13_1)
Type	Single-tuned	Single-tuned
Capacitor rating	6500.9 kvar (1-phase)	21228.5 kvar (1-phase)
Capacitance	1.188 μF	3.878 μF
Rated voltage	132 kV	132 kV
Inductor reactance X_{L1}	22.1506 Ω	6.7833 Ω
Quality factor	20	20

Four kinds of study were run on this model: a Newton-Raphson load flow for the voltage profile and the loss figures, a harmonic load flow for the VTHD and VIHD reports and the waveform and spectrum plots, a short circuit study to get the fault levels used for sizing the SVCs, and a contingency study in which the HVDC link and its filters were switched out automatically.

4 Implementation

4.1 Description

This section describes the step-by-step implementation of the extended power system studied in this project. Beginning with the standard IEEE 30-bus test system, the network was progressively built up to incorporate a new remote generation source (Bus 31) situated 750 km from the existing grid, with two alternative transmission technologies — High Voltage AC (HVAC) and High Voltage DC (HVDC) — implemented and refined to deliver this power reliably to the main network. Beyond the base load flow implementation, harmonic analysis and filter design were carried out to address the characteristic harmonics introduced by the HVDC converter stations, and contingency analysis was performed to assess the system's ability to withstand credible outage conditions. The following subsections document each stage of this process in sequence: verification of the base network, extension of the system with the new bus and remote generation, implementation of the HVAC transmission link, implementation of the HVDC transmission link, harmonic filter implementation, and contingency analysis setup. The corresponding results, comparative analysis, and interpretation of these implementations are presented separately in section 5.3.

4.2 Base Case Verification

Before any modification to the network, the standard IEEE 30-bus test system was first constructed in ETAP 16.0.0 and a base-case Load Flow study was run to confirm that the model behaved correctly and converged reliably. This step was essential to establish a trustworthy reference point: any effects observed later from the Bus 31 extension could then be attributed with confidence to the new addition itself, rather than to an underlying modeling error in the original network. The base case converged successfully and matched expected behavior for the standard test system, and was taken as the verified starting point for all subsequent additions described in the following subsections. The full base-case results are presented and analyzed in Section 5.3.

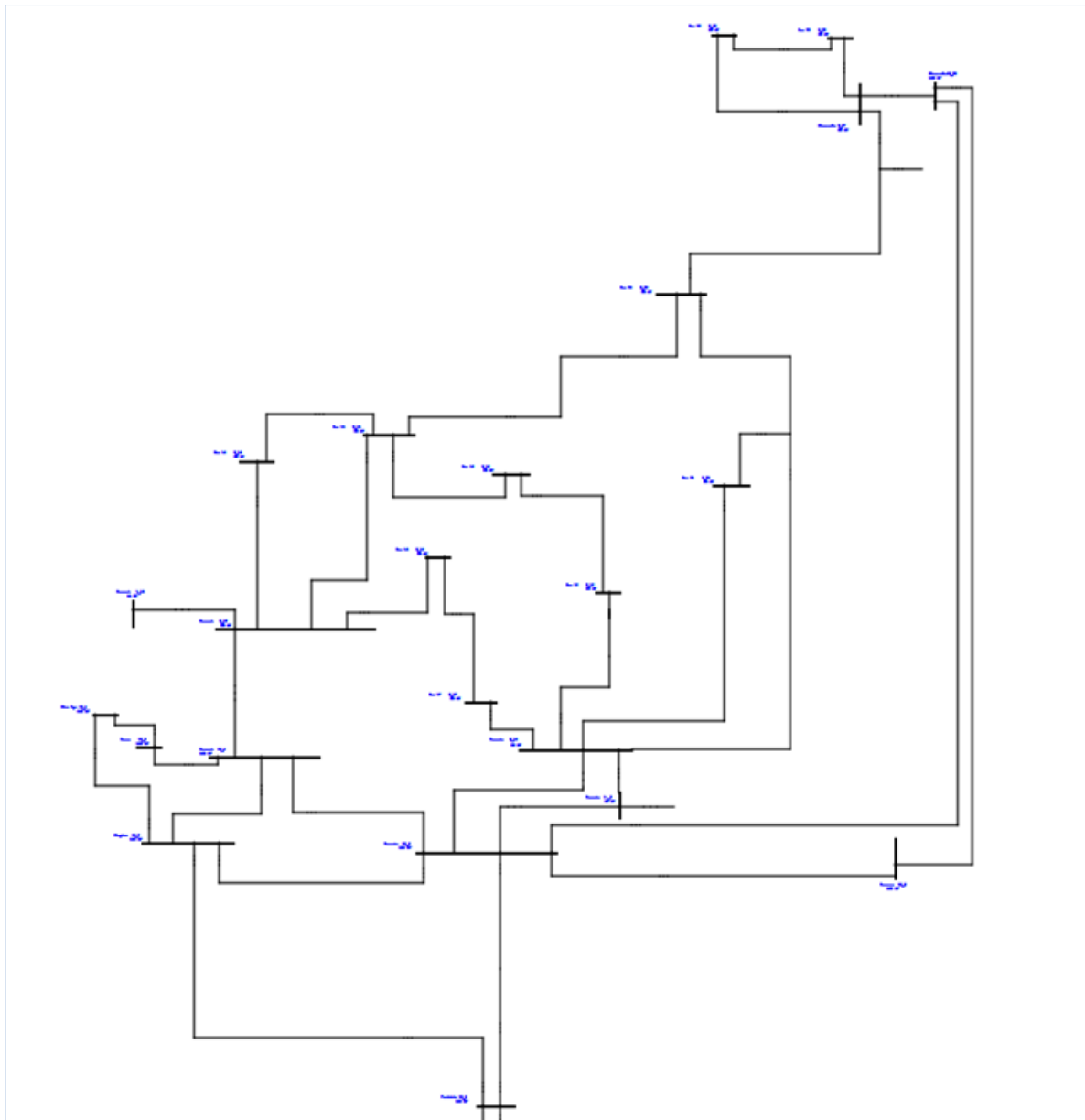


Figure 4: One-line diagram of the standard IEEE 30 – Bus system as implemented in ETAP, prior to the Bus 31 extension.

4.3 System Input Data – IEEE 30 – Bus System

Table 5: Bus Nominal Voltage and Load Data

Bus	ETAP ID	Nominal kV	Load (MW)	Load (Mvar)
1	Glen Lyn 13_1	132	—	—
2	Claytor 13_2	132	21.7	12.7
3	Kumis 13_3	132	2.4	1.2
4	Hancock 13_4	132	7.6	1.6
5	Fieldale 13_5	132	94.2	19.0
6	Roanoke 13_6	132	—	—
7	Blaine 13_7	132	22.8	10.9
8	Reusens 13_8	132	30.0	30.0
9	Roanoke 1_9	33	—	—
10	Roanoke 3_10	33	5.8	2.0
11	Roanoke 1_11	11	—	—
12	Hancock 3_12	33	11.2	7.5
13	Hancock 1_13	11	—	—
14	Bus 14 3_14	33	6.2	1.6
15	Bus 15 3_15	33	8.2	2.5
16	Bus 16 3_16	33	3.5	1.8
17	Bus 17 3_17	33	9.0	5.8
18	Bus 18 3_18	33	3.2	0.9
19	Bus 19 3_19	33	9.5	3.4
20	Bus 20 3_20	33	2.2	0.7
21	Bus 21 3_21	33	17.5	11.2
22	Bus 22 3_22	33	—	—
23	Bus 23 3_23	33	3.2	1.6
24	Bus 24 3_24	33	8.7	6.7
25	Bus 25 3_25	33	—	—
26	Bus 26 3_26	33	3.5	2.3
27	Cloverdle 3_27	33	—	—
28	Cloverdle13_28	132	—	—
29	Bus 29 3_29	33	2.4	0.9
30	Bus 30 3_30	33	10.6	1.9
Total (30 buses)			283.4	126.2

Shunt capacitors (constant-Z elements): 19.0 MVAR at Bus 10 and 4.3 MVAR at Bus 24, totaling 23.3 MVAR.

Table 6: Generators Input Data

Bus	ETAP ID	kV	Mode	MW	MVAR max	MVAR min	V setpoint (%)
1	Glen Lyn 13_1	132	Swing	—	—	—	103.0
2	Claytor 13_2	132	Voltage control	40.0	50.0	−40.0	101.0
5	Fieldale 13_5	132	Voltage control	25.0	40.0	−29.4	104.0
8	Reusens 13_8	132	Voltage control	30.0	50.0	−10.0	103.0
11	Roanoke 1_11	11	Voltage control	15.0	30.0	−6.0	101.0
13	Hancock 1_13	11	Voltage control	15.0	10.6	−6.0	101.0
Total (excl. swing)				125.0			

Bus 1 (Glen Lyn) is the swing bus; its real and reactive output is determined by the load flow solution and is therefore not specified as input. Other necessary system information is provided in

the appendix.

4.4 System Extension – Bus 31 and Remote Generation

The base IEEE 30-bus network described in section 4.2 no long transmission corridor, and therefore offers no natural point at which HVAC and HVDC transmission can be compared. To create one, the network was extended with a remote generation complex connected to the existing system through a single 750km link. Because both link technologies were later modelled between the same pair of terminal buses, any difference in losses, voltage profile or reactive requirement can be attributed to the transmission technology itself rather than to a difference in the surrounding network.

Four new buses were added. At the sending end, GEN31_bus (13.8 kV) serves as the generator terminal bus, Bus31_400 (400 kV) as the transmission-voltage bus from which the link originates, and Bus31 (132 kV) as a local distribution bus supplying load in the vicinity of the new plant. At the receiving end, BUS1_400 (400 kV) was introduced adjacent to the existing Glen Lyn 132 kV bus to terminate the link.

The new generating unit G31 was modelled at GEN31_bus with a rating of 149.714 MW at 13.8 kV and a lower reactive limit of -80 MVAR, the negative limit being required so that the machine can absorb reactive power during light-load and HVAC-charging conditions. It is connected through the generator step-up transformer GSU_31, rated 150 MVA, to Bus31 at 132 kV, which acts as the plant's collector bus and also supplies a local load of 21.051 MVA. A second transformer, T31, rated 200 MVA, then steps 132 kV up to the 400 kV transmission level at Bus31_400, from which the link to the existing network originates. At the receiving end, BUS1_400 is tied into the existing network through T1_HVAC, rated 200 MVA, which interfaces the 400 kV terminal with the 132 kV Glen Lyn bus. This receiving-end arrangement is common to both the HVAC and the HVDC configurations, so that only the link itself changes between the two studies.

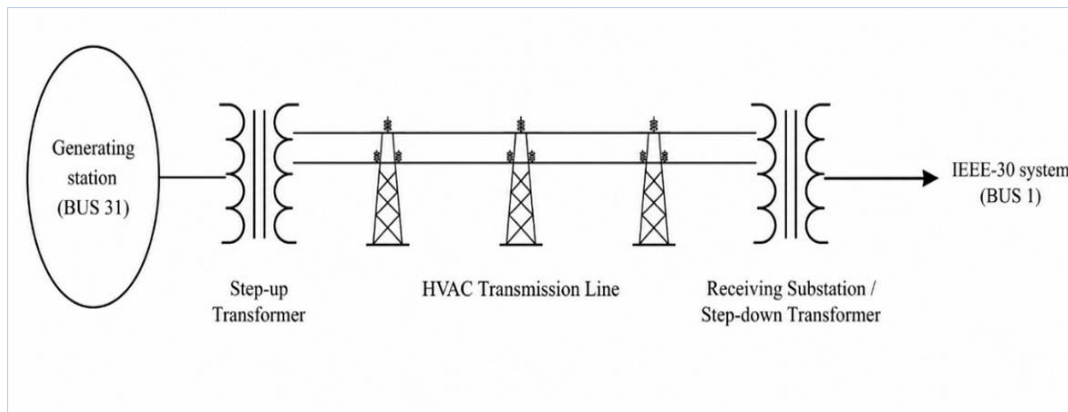


Figure 5: HVAC Transmission Line

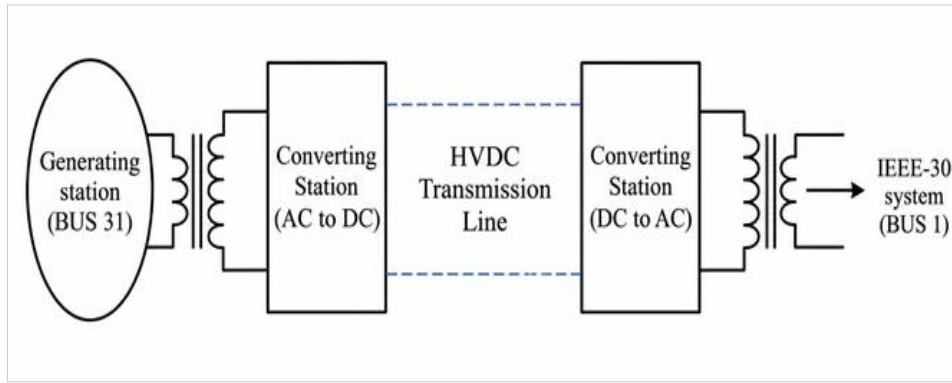


Figure 6: HVDC Transmission Model

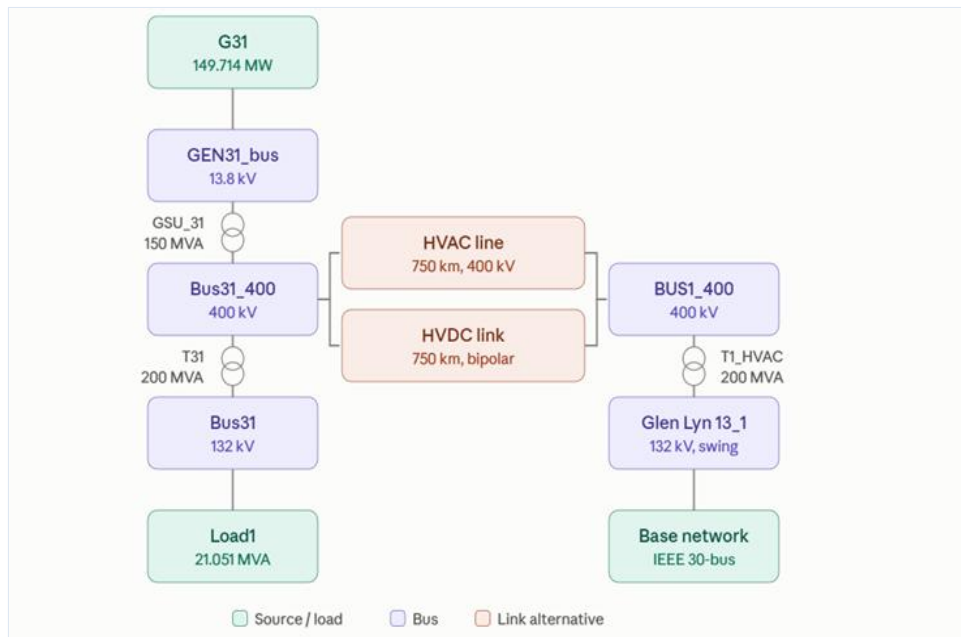


Figure 7: Single-line diagram of the extension added to the IEEE 30-bus system, showing the new generation at Bus 31 and the 750 km link to the existing network at Glen Lyn.

Table 7: Equipment Added to the Base System

Element	ID	Connection	Rating
Bus	GEN31_bus	—	13.8 kV
Bus	Bus31_400	—	400 kV
Bus	Bus31	—	132 kV
Bus	BUS1_400	—	400 kV
Generator	G31	GEN31_bus	149.714 MW, 13.8 kV, $Q_{\min} = -80\text{MVAR}$
Transformer	GSU_31	GEN31_bus – Bus31_400	150 MVA
Transformer	T31	Bus31_400 – Bus31	200 MVA
Transformer	T1_HVAC	BUS1_400 – Glen Lyn 13_1	200 MVA
Load	Load1	Bus31	21.051 MVA

All elements were placed on the one-line diagram from the AC edit toolbar and their ratings entered through the respective element editors, with the system base frequency retained at 50 Hz. The swing

bus assignment was left unchanged at Glen Lyn 13_1, so that the new plant operates as a scheduled-output source and the existing slack machine absorbs the resulting mismatch. No parameter of the original 30-bus network was modified during this step. The extended system therefore comprises 34 buses and 7 generators, and forms the common platform on which the HVAC link (Section 4.5) and the HVDC link (Section 4.6) were subsequently modelled.

4.5 HVAC Transmission Line Implementation

The first of the two transmission alternatives was implemented as a transmission line element, HVAC Line drawn between Bus31_400 and BUS1_400. Both terminals operate at 400 kV, and the line was defined as a three-phase circuit in the In-service, As-Built state with a length of 750 km and zero length tolerance. A single circuit was used rather than a double-circuit arrangement, so that the transfer capability of one HVAC corridor could be compared directly against one HVDC link of the same route length.

The line was represented by its per-unit-length sequence impedance rather than by conductor geometry. In the Impedance tab the entry mode was set to User-Defined and the positive-sequence values were entered directly, in ohms per kilometer, at the project frequency of 50 Hz. As a result the Parameter tab (conductor type, GMR, outside diameter) and the Configuration tab (tower geometry, phase spacing, ground wires) were left at their default zero values, since ETAP uses those fields only when the impedance is set to Calculated. This choice was made because the study is concerned with the electrical behavior of a corridor of representative parameters, not with the design of a particular tower and conductor.

Table 8: HVAC Line input Parameter

Parameter	Value
From – To	Bus31_400 – BUS1_400
Nominal voltage	400 kV
Length	750 km
Connection	3-phase, 1 conductor per phase
Positive-sequence R (T1 = 25 °C)	0.036 Ω /km
Positive-sequence R (T2 = 75 °C)	0.036 Ω /km
Positive-sequence X	0.31 Ω /km
Positive-sequence Y	3.77 μ S/km
Operating temperature	75 °C (min and max)
Frequency	50 Hz

Since the same resistance was entered at both the 25 °C and 75 °C library temperatures, no temperature correction is applied and the line resistance stays at 0.036 Ω /km throughout the study, with the operating temperature fixed at 75 °C. Over the full 750 km this gives a total series impedance of 27 Ω resistance and 232.5 Ω reactance, and a total shunt admittance of 2.83 mS.

That shunt admittance is what makes the HVAC case demanding. At 400 kV the distributed capacitance of the line generates approximately 452 MVAR of charging reactive power at no load, which is why fixed shunt reactors were required at both 400 kV terminals for the load flow to converge within acceptable voltage limits.

Table 9: Shunt Reactors HVAC Compensated Case

Reactor ID	Connected bus	Nominal voltage	Rating
shunt31	Bus31_400	400 kV	170 MVA
shunt1_400	BUS1_400	400 kV	250 MVA
shunt1	Glen Lyn 13_1	132 kV	30 MVA

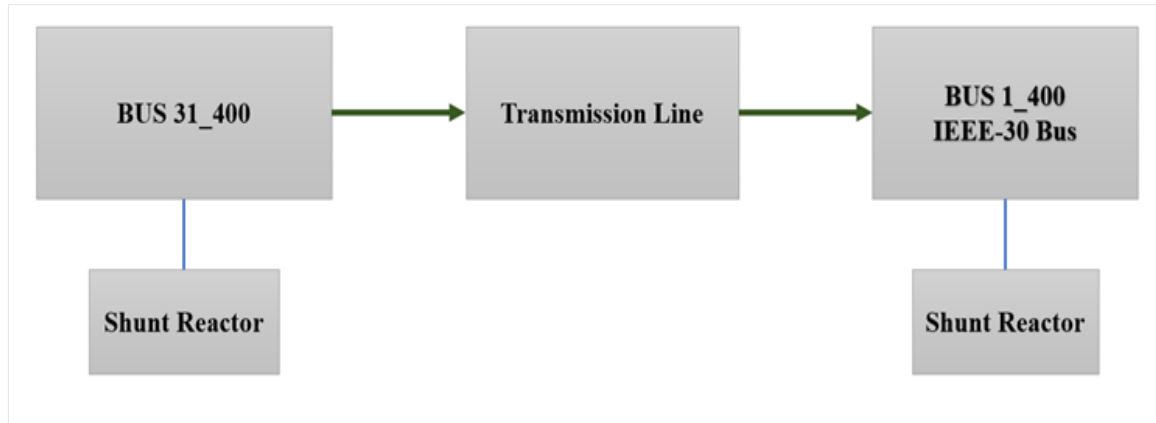


Figure 8: Block representation of the HVAC transmission link. The 750 km, 400 kV line connects the new generation at Bus31_400 to BUS1_400 at the IEEE 30-bus network, with fixed shunt reactors at both terminals to absorb the line charging reactive power

4.6 HVDC Transmission Implementation

The second transmission alternative was implemented using ETAP's DC Link element, DC_LINK1, connected between the same pair of terminals as the HVAC line, from Bus31_400 to the BUS1_400, both at 400 kV. The element was placed in the In-service, As-Built state. Because DC_LINK1 and LineHVAC occupy the same corridor, only one of the two is kept in service in any given study case, and the corridor length of 750 km is represented in the DC link through its conductor resistance rather than through a length field.

The link is a point-to-point bipolar scheme with four six-pulse bridges, giving two bridges per pole in series. Each end is interfaced to the 400 kV AC system through a converter transformer rated 150 MVA with a primary voltage of 400 kV, a secondary voltage of 132 kV, a commutating reactance of 12% and a tap setting of 5%. The rectifier and inverter transformers were given identical ratings, so the scheme is symmetrical and can be examined in either direction of power transfer without changing the equipment data. Both AC systems operate at 50 Hz.

The DC circuit itself was specified by a total conductor resistance of 54 Ω , a rated direct voltage of 660 kV across the poles (± 330 kV to ground) and a rated direct current of 0.17 kA, which corresponds to a rated transmitted power of 112.2 MW. A maximum current limit of 150% of rating was retained for overload studies. The firing and extinction angles were constrained to a minimum alpha of 5° and a maximum of 20° at the rectifier, and a minimum gamma of 15° and a maximum of 20° at the inverter, the gamma floor being the usual commutation margin required to avoid commutation failure at the inverter.

Table 10: DC Link Input Parameter

Parameter	Value
From – To	Bus31_400 – BUS1_400
AC terminal voltage, both ends	400 kV, 50 Hz
Configuration	Bipolar, 4 bridges
Converter transformer, primary / secondary	400 kV / 132 kV
Converter transformer MVA, X _c , tap	150 MVA, 12%, 5%
DC conductor resistance	54 Ω
Rated V _{dc} / I _{dc} / P _{dc}	660 kV / 0.17 kA / 112.2 MW
Maximum current (I _{max})	150% of rating
Alpha, minimum / maximum	5° / 20°
Gamma, minimum / maximum	15° / 20°

With these settings the load flow converges to an operating point of 655.9 kV and 0.153 kA, transmitting 100.353 MW, with the rectifier firing angle at 5° and the inverter extinction angle at 17°. The direct voltage therefore sits close to its rated value while the current runs below rating, and the extinction angle settles comfortably inside its permitted band.

Because line-commutated converters consume reactive power at both terminals in proportion to the transmitted power, local reactive support was provided rather than drawing it through the AC network. A static VAR compensator rated 20 MVA, SVC31_400, was connected at the rectifier terminal Bus31_400, and a second rated 45 MVA, SVC_gen_1, at Glen Lyn 13_1 on the 132 kV side of the inverter terminal. Static VAR compensators were preferred to fixed capacitor banks because the converter reactive demand varies with transmitted power, and a bank sized for rated transfer would overcompensate and raise the terminal voltage when the link operates below rating, as it does in the base case.

Table 11: Reactive Compensation in HVDC Configuration

Element	Bus	Nominal voltage	Rating
SVC31_400	Bus31_400	400 kV	20 MVA
SVC_gen_1	Glen Lyn 13_1	132 kV	45 MVA

4.7 Harmonic Analysis and Filter Design

The integration of a bipolar HVDC link utilizing line-commutated converters inherently introduces harmonic currents into the AC network. Specifically, a 12-pulse converter injects characteristic harmonics of order $h = 12k \pm 1$, making the 11th and 13th harmonics dominant in the system. Initial harmonic load flow studies revealed that these injections drove the voltage distortion far above permissible limits, reaching a Total Harmonic Distortion (THD) of 5.56% at major buses. Unmitigated, this level of distortion causes excessive transformer heating, potential nuisance relay trips, and general power quality degradation for stakeholders connected to the grid. Passive harmonic filters were implemented to shunt these specific frequencies to ground and restore system compliance

Determination of Filter Specifications To mitigate the distortion, two single-tuned passive filters (configured as series R-L-C branches) were designed. The design process calculated the exact component specifications based on both the reactive power requirements of the system and the target harmonic frequency. The mathematical formulation was executed as follows:

- **Capacitor Sizing:** The capacitor was selected first based on the reactive power (Q_f) it needed to supply to the network, using the relationship $X_{C1} \approx \frac{V_{LL}^2}{Q_f}$.
- **Inductor Tuning:** The filter was tuned to the target harmonic order n by equating the inductive and capacitive reactances at the resonant frequency ($X_L(\omega_n) = X_C(\omega_n)$). This yielded the required inductive reactance: $X_L = \frac{X_C}{n^2}$.
- **Damping Resistance:** To achieve a sharp tuning profile, a quality factor (q) of 20 was selected. The necessary damping resistance was calculated from the characteristic impedance ($X_0 = \sqrt{\frac{L}{C}}$) as $R = \frac{X_0}{q}$.

applying these formulas with a selected quality factor of 20 yielded the following final hardware specifications for the two filters:

Table 12: Specification of The Harmonic Filter

Specification	Filter 1 (HF7)	Filter 2 (HF3)
Connected Bus	Bus 31	Bus 30 (Glen Lyn 13_1)
Rated Voltage	132 kV	132 kV
Capacitor Rating	6500.9 kvar	21228.5 kvar
Capacitance	1.188 μF	3.878 μF
Inductor Reactance	22.1506 Ω	6.7833 Ω
Quality Factor (q)	20	20

4.8 Contingency Analysis

To check the system's stability in case the HVDC link gets disconnected from system, we performed a contingency test. In real power systems, HVDC AC harmonic filters are connected to the AC busbar using conventional AC circuit breakers (often SF6 gas circuit breakers or high-voltage alternating current circuit breakers) (Csanyi, 2020) ^[8]. These standard AC breakers automatically switch the filter banks on and off to manage changing reactive power levels and harmonic limits on the AC grid. So as soon as HVDC link will get disconnected from the grid, the filters will be switched off.

5 Design Analysis and Evaluation

5.1 Novelty

The novelty of this project is not a new device. It is in how the study was set up and carried through. Six points are worth noting.

- **A fair comparison instead of an easy one.** Most published studies compare an HVDC link against an AC line that has no compensation at all. That AC line collapses, so HVDC looks like a huge winner. In this project the HVAC option was first repaired properly, with 420 MVAR of shunt reactors and a 30 MVAR SVC, until it had no critical alarm at all. Only then were the two options compared. The result is a much smaller and much more honest margin: 1.543 MW against 1.264 MW of link loss and 11.072 MW against 10.817 MW of total system loss.
- **The comparison is tied back to break-even theory.** The small margin is not treated as a disappointing result but as a confirmation. At 750 km the corridor sits close to the usual break-even distance for overhead lines, so a small margin is exactly what theory predicts. The project therefore shows where HVDC stops being obviously better, which is more useful than one more paper saying HVDC is best.
- **Three different studies on one network.** Load flow, harmonic analysis and contingency analysis are usually done in separate papers on separate systems. Here all three were done on the same modified IEEE 30-bus model, so the interaction between them is visible. For example, the filters designed for the harmonic problem also supply reactive power, and when the link and its filters are lost together in the contingency study, the network loses both the power flow and that reactive support at the same time.
- **The contingency is the loss of the link itself.** Contingency studies normally remove one AC line. Removing the whole HVDC link together with its filters is a more severe and more realistic event for a system that depends on a single DC corridor, and it is the case that revealed the under-voltage at Buses 26, 29 and 30.
- **Equipment ratings come from calculations, not from guessing.** The shunt reactor size came from the calculated 452 MVAR of line charging power, the filter parameters came from the single-tuned filter equations. Every number in the design can be traced back to a formula.
- **A practical extra finding.** The filters were designed for the highest characteristic harmonic that was violating the limit, but in the simulation, they reduced both the 11th and the 13th order harmonics, from about 4% down to about 0.5%. Two filters were therefore enough where four might have been expected.

5.2 Design Considerations (PO(c))

The design was checked against the safety, environmental and social needs described below. Because this project is a simulation study, these considerations were applied to the equipment that the design asks to be built: the 750 km corridor, the converter stations, the shunt reactors, the SVCs and the two filter banks.

5.2.1 Considerations to public health and safety

Voltage limits protect people and equipment. The uncompensated HVAC case had 838 kV on a 400 kV bus, more than twice the rated value. Such a voltage would flash over insulators and could destroy a transformer, which is a direct danger to anyone working near it. Keeping every bus between 0.95 and 1.05 per unit was therefore treated as a safety requirement, not just a quality requirement.

Equipment overload. T1_HVAC was loaded to 389.2% in the uncompensated case. An overloaded transformer overheats, its oil degrades and it can fail violently. The compensation was sized so that no transformer, generator or line runs above its rating.

Harmonic distortion. Harmonics cause extra heating in transformers, motors and cables, they can make protection relays operate wrongly and they can overheat capacitor banks. Bringing the THD down to the IEEE Std 519 range is a safety measure as much as a power quality measure.

Filter capacitor safety. The filter capacitors store energy and stay charged after the bank is switched off. In a real installation they need discharge resistors, an interlocked and fenced enclosure, clear warning signs and a locked earthing switch before anyone touches them.

DC specific hazards. A DC arc does not have a natural current zero, so it does not extinguish on its own. DC switching therefore needs special breakers and strict procedures. The converter valve hall also needs interlocks and its cooling and fire protection systems must be maintained.

Clearance and electric field. A 400 kV AC line and a 660 kV DC line both need large ground clearance and a controlled right of way so that the electric field at ground level and the risk of contact stay within safe values.

5.2.2 Considerations to environment

Less corona, less noise and less radio interference. Corona is the ionization of the air immediately around a conductor, which happens when the electric field at the surface exceeds about 30 kV/cm. It produces a crackling noise, radio interference, ozone and a small power loss. This phenomenon is proportional to the rate of change of current in the conductor. As DC current don't change over time it has lower corona effect. So, people and wildlife living near a DC corridor are less affected.

Lower loss means less fuel and less carbon dioxide. The HVDC option saves about 0.255 MW of loss compared with the compensated HVAC option (10.817 MW against 11.072 MW). Running continuously for a year this is roughly 2.2 GWh of energy that does not have to be generated. If that energy would have come from a gas fired plant, the saving is on the order of a thousand tons of carbon dioxide a year. The harmonic filters help in the same way, because harmonic currents produce heat and no useful work.

Narrower right of way. A bipolar DC line needs two conductors instead of three and its towers can be smaller for the same power. Less land is cleared, fewer trees are cut and less farmland is taken.

Fewer materials over the life of the line. Two conductors instead of three means less aluminum and steel per kilometer over 750 km.

The other side of the picture. The converter stations occupy land, contain large amounts of copper, oil and cooling equipment, and use insulating gas in the switchgear. Any leakage of SF₆ is a very strong greenhouse gas. The capacitors and reactors also become electronic waste at the end of their life and must be recycled properly. These costs were noted, but they were not quantified in this project.

5.2.3 Considerations to cultural and societal needs

Bringing remote energy to where people live. The whole point of the study is to make a 150 MW plant that is 750 km away useful to the grid. In a country such as Bangladesh this matches the real need to bring power from remote sites, from offshore areas or from a neighboring country to the load centers near the cities.

Less land taken from the community. A narrower right of way means fewer families displaced, less farmland lost and fewer religious, cultural or heritage sites disturbed along a 750 km route. Land acquisition is usually the most sensitive social issue in a long transmission project.

Cheaper and steadier electricity. Lower losses mean less fuel burnt for the same delivered energy, which helps to keep the tariff down. Better voltage and lower harmonic distortion also mean fewer equipment failures for industrial and domestic customers.

Reliability for the people at the far end. The contingency study exists for a social reason. If the whole area depends on one DC link, then losing that link must not black out the Buses. Our study shows that no such phenomenon occurs.

Local skills and jobs. A converter station needs trained operators and maintenance staff, which creates skilled local employment, but it also means that training must be arranged before the station is commissioned.

Visual and cultural acceptance. Towers are visible for a long way. Route selection should avoid sites of religious or cultural importance, and the community should be consulted before the route is fixed.

5.3 Investigations (PO(d))

5.3.1 Design of Experiment

purpose of the experiment was to compare high voltage AC and high voltage DC transmission over a long corridor, using transmission losses and steady-state voltage profile as the basis of comparison. The extended system described in Chapter 4 was constructed so that the two technologies could be interchanged between a fixed pair of terminals, Bus31_400 and BUS1_400, without altering anything else in the network.

Two study cases were defined. In the first, the corridor was represented by the HVAC transmission line Line-HVAC, with shunt compensation added at the terminals until the terminal voltages settled within the acceptable band. In the second, Line-HVAC was taken out of service and replaced by the DC link DC_LINK1 between the same two buses, with static VAR compensation provided at the converter terminals on the same basis. Each case was therefore a compensated, converged operating point rather than an uncompensated one, so that the comparison reflects transmission schemes as they would actually be built.

The independent variable was the transmission technology in the corridor. The dependent variables were the branch losses in the corridor and the bus voltage magnitudes throughout the extended system. Held constant across both cases were the terminal buses and their 400 kV nominal voltage, the corridor length of 750 km, the generation at G31, the local load at Bus31, the receiving-end interface transformer T1_HVAC, and the entire original IEEE 30-

bus network, whose parameters were not modified at any point. The reactive compensation was necessarily different between the two cases, since the HVAC line requires absorption of its charging reactive power while the converters require a supply of reactive power; the compensation was therefore treated not as a variable under test but as a part of each scheme sized to bring that scheme to an acceptable operating point.

A Newton-Raphson load flow was run for each case in ETAP, and the resulting branch loss and bus voltage reports form the basis of the results presented in the following subsections.

5.3.2 Data Collection

Data was collected from ETAP's Load Flow Report Manager after each study case had converged. A separate output set was generated for each stage of the study, named according to the configuration in service, and saved as PDF reports in the project directory so that the cases could be compared without re-running any simulation.

For each case, two reports were taken from the Summary tab. The Losses report provided the active and reactive power loss in every branch, from which the loss in the transmission corridor and the total system loss were extracted. The Bus Loading report provided the voltage magnitude and angle at every bus, which formed the voltage profile. The Alert reports were also generated for each case, recording any bus voltages or branch loadings outside their marginal and critical thresholds; these were used both to identify the deficiencies of the uncompensated configurations and to confirm that the compensated configurations operated within limits.

The quantities recorded for each case were therefore the corridor loss, the total system real and reactive loss, the voltage magnitude at every bus of the extended system, and any alerts raised. These are presented in the following subsection's Tables.

5.3.3 Results and Analysis

Step – 1 : Original IEEE 30 – Bus System

The base case converged in 3 iterations with zero mismatch. Total losses were 8.997 MW and 3.490 Mvar against a demand of 292.399 MW, or 3.08%. Voltages ranged from 95.987% at Bus 30 to 103.000% at the swing bus.

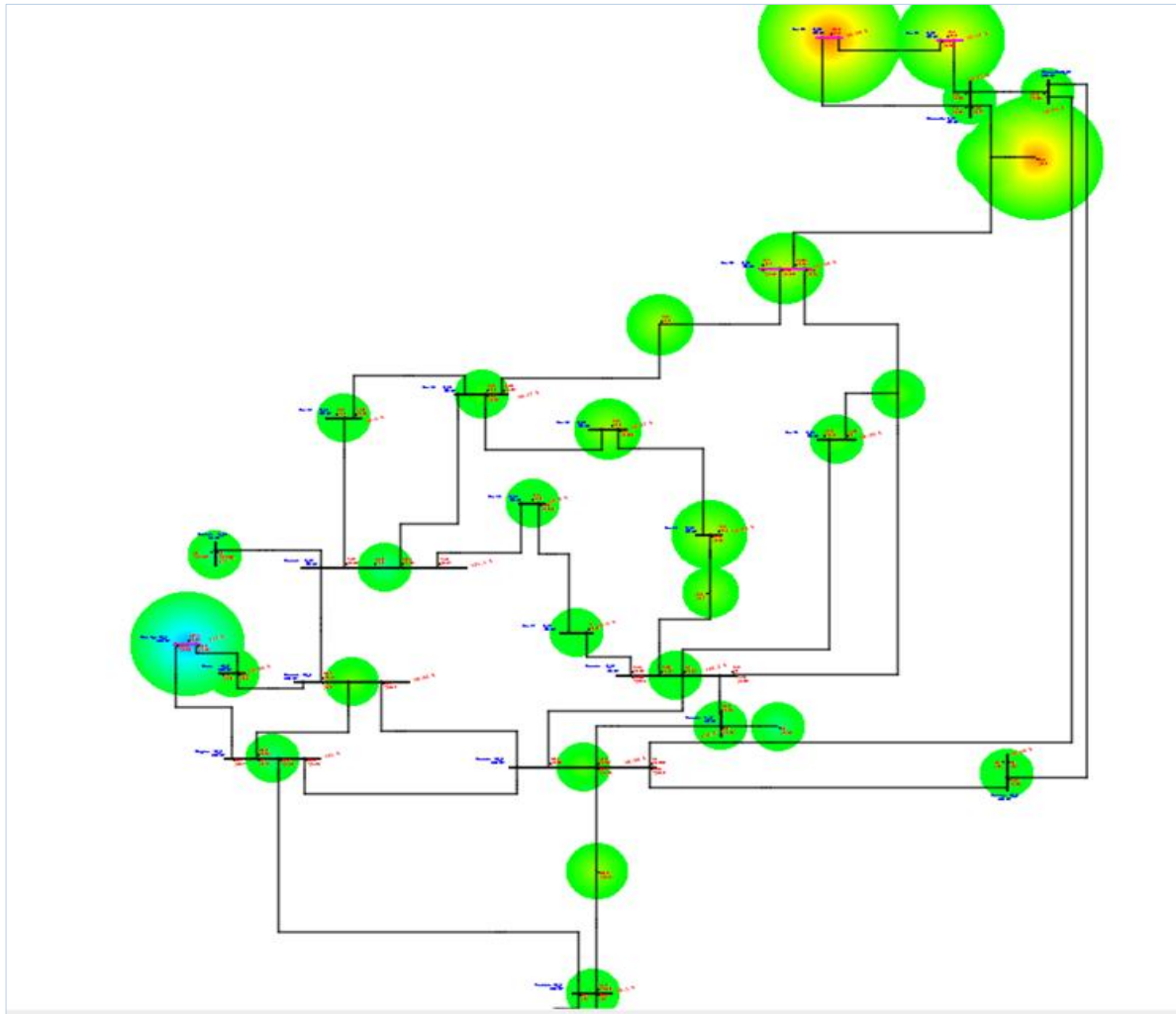


Figure 9: Load flow voltage profile of the original IEEE 30-bus system. Green markers indicate buses within limits; the larger warm-colored markers identify the buses operating at the extremes of the profile.

Table 13: Generation Demand and Losses in Base Case

Quantity	MW	Mvar	MVA	% PF
Source (swing bus)	167.399	5.025	167.475	99.95
Source (non-swing)	125.000	101.459	160.994	77.64
Total demand	292.399	106.485	311.185	93.96
Total static load	283.400	126.200	—	—
Shunt capacitors	0.002	-23.205	—	—
Total losses	8.997	3.490	—	—

Table 14: Alerts - Base Case

Device	Condition	Limit	Operating	%
Gen_5	Over excited	40.000 Mvar	40.000	100.0
Gen_8	Over excited	50.000 Mvar	50.00	100.0
Bus 24	Under voltage	33.000 kV	32.319	97.9
Bus 26	Under voltage	33.000 kV	31.79	96.3
Bus 29	Under voltage	33.000 kV	32.07	97.2
Bus 30	Under voltage	33.000 kV	31.68	96.0
Glen Lyn 13_1	Over voltage	132.000 kV	135.96	103.0

Alert thresholds were left at the ETAP defaults: 98% marginal and 95% critical for undervoltage, 102% and 105% for overvoltage.

Gen_5 and Gen_8 are flagged at 100% of their reactive limits. This is not an overload; both machines are simply delivering the reactive power they are rated to deliver, and the system is operating normally. It does mean, however, that neither retains any reactive margin, so neither can support its bus voltage further if demand increases or a branch is lost.

That absence of margin is reflected in the voltage profile. The four undervoltage buses are all in the 33 kV radial group fed from Cloverdle — Buses 24, 26, 29 and 30 — which is the part of the network furthest from generation and dependent on reactive support arriving through several transformations. Bus 30, at the end of that group, is the weakest bus in the system. The overvoltage at Glen Lyn 13_1 is a consequence of the swing bus being scheduled at 103%, not a system deficiency.

All alerts are marginal rather than critical, so the base system is acceptable as a reference. Its weak point is reactive support at the 33 kV periphery, and that is the condition against which the effect of integrating the new generation at Bus 31 is assessed in the following parts.

Step – 2 : HVAC Implementation – Uncompensated

The 750 km HVAC line was placed in service between Bus31_400 and BUS1_400 with G31 dispatched. The load flow converged in 3 iterations, but to an operating point far outside acceptable limits.

Table 15: Critical Alerts HVAC integration

Device	Condition	Limit	Operating	%
Bus31_400	Over voltage	400.000 kV	838.22	209.6
Bus31	Over voltage	132.000 kV	272.39	206.4
GEN31_bus	Over voltage	13.800 kV	28.14	203.9
BUS1_400	Over voltage	400.000 kV	593.285	148.3
G31	Under excited	–80.000 Mvar	–80.00	—
Gen_1	Under excited	16.1 Mvar(peak)	–770.82	—
Gen_5	Over excited	40.000 Mvar	40.00	100.0
Gen_8	Over excited	50.000 Mvar	50.00	100.0
T1_HVAC	Overload	200.000 MVA	778.49	389.2

Table 16: Marginal alerts — HVAC integration

Device	Condition	Limit	Operating	%
Bus 24	Under voltage	33.000 kV	32.320	97.9
Bus 26	Under voltage	33.000 kV	31.80	96.3
Bus 29	Under voltage	33.000 kV	32.07	97.2
Bus 30	Under voltage	33.000 kV	31.677	96.0
Glen Lyn 13_1	Over voltage	132.000 kV	135.96	103.0
GSU_31	Overload	150.000 MVA	144.22	96.1

The result is a reactive power surplus, not a deficit. Total reactive losses are -884.124 Mvar, almost all of it from the line: Line-HVAC alone contributes -1234 Mvar, the charging power of 750 km of 400 kV line rising with the square of the terminal voltage.

No machine can absorb it. G31 is driven to its lower limit of -80 Mvar, and Gen_1, whose reactive capability is 16.1 Mvar, is placed at -770.822 Mvar — some forty-eight times that value. Both machines therefore lose voltage control, and the voltage rises unchecked across the extension: 209.6% at Bus31_400, 206.4% at Bus31, 203.9% at GEN31_bus and 148.3% at BUS1_400. A converged load flow is not a feasible operating point here, since a real machine would lose excitation control long before this condition; the voltages indicate the severity of the reactive imbalance rather than predicting actual bus values.

The transformers carry the same imbalance. T1_HVAC is loaded to 778.49 MVA against its 200 MVA rating, at 389.2%, and GSU_31 reaches 96.1% of rating. Within the original network the under voltages at Buses 24, 26, 29 and 30 are unchanged from the base case, so the new generation brings no benefit to the weak part of the network in this state.

Compensation is therefore essential. Shunt reactors must absorb the line's charging power at both 400 kV terminals, sized to return both machines within their reactive limits and restore voltage control.

Step – 3: HVAC Integration with Compensation

Three shunt reactors were added: shunt31 rated 170 MVA at Bus31_400, shunt1_400 rated 250 MVA at BUS1_400, and shunt1 rated 30 MVA at Glen Lyn 13_1. The load flow converged in 3 iterations.

Table 17: Effect of compensation — HVAC case

Quantity	Uncompensated	Compensated
Bus31_400	209.6%	104.0%
Bus31	206.4%	101.6%
GEN31_bus	203.9%	100.0%
BUS1_400	148.3%	101.5%
Gen_1 reactive output	-770.82 Mvar	6.883 Mvar
G31 reactive output	-80.000 Mvar (at limit)	-24.712 Mvar
T1_HVAC loading	389.2%	50.9%
GSU_31 loading	96.1%	83.0%

Table 18: Alerts — HVAC compensated

Device	Condition	Limit	Operating	%
Gen_5	Over excited (critical)	40.000 Mvar	40.000	100.0
Gen_8	Over excited (critical)	50.000 Mvar	50.00	100.0
Bus 24	Under voltage	33.000 kV	32.320	97.9
Bus 26	Under voltage	33.000 kV	31.80	96.3
Bus 29	Under voltage	33.000 kV	32.07	97.2
Bus 30	Under voltage	33.000 kV	31.68	96.0
Bus31_400	Over voltage	400.000 kV	415.96	104.0
Glen Lyn 13_1	Over voltage	132.000 kV	135.96	103.0

Table 19: HVAC transmission line operating parameters

Parameter	Sending end Bus31_400	Receiving end BUS1_400
Active power, P	98.645 MW	97.101 MW
Reactive power, Q	-232.001 Mvar	-232.540 Mvar
Apparent power, S	252.102 MVA	251.999 MVA
Line-to-line voltage	416.0 kV (104.0%)	406.0 kV (101.5%)
Line current, I	349.9 A	358.4 A
Active line loss, P_loss	1.5433 MW	

Table 20: Shunt reactors — HVAC compensated case

Reactor ID	Connected bus	Nominal voltage	Rating
shunt31	Bus31_400	400 kV	170 MVA
shunt1_400	BUS1_400	400 kV	250 MVA
shunt1	Glen Lyn 13_1	132 kV	30 MVA

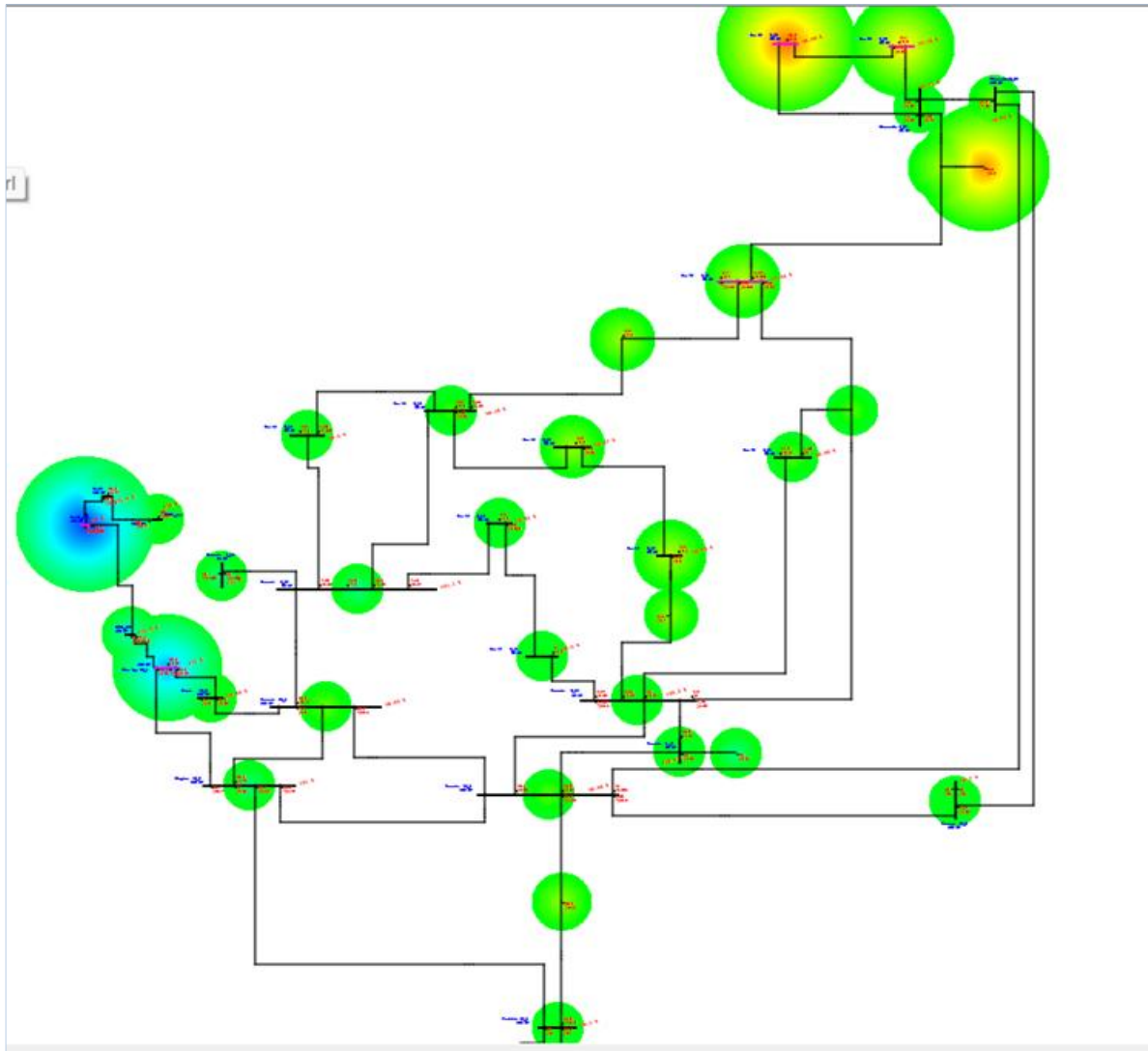


Figure 10: Load flow voltage profile of the HVAC compensated system. Green markers indicate buses within limits; the larger warm-colored markers identify the buses operating at the extremes of the profile

Analysis of HVAC Integration's Result

The compensation restores a workable operating point. Reactive losses fall from -884.124Mvar to -439.709Mvar , and the reactors absorb the remainder locally rather than pushing it into the machines: Line-HVAC still generates -464.5Mvar , but this is now taken by the shunt reactors instead of the generators.

Both machines regain voltage control. Gen_1 returns to 6.883Mvar , within its 16.1Mvar capability, and G31 to -24.712Mvar , comfortably inside its -80Mvar limit. With excitation control recovered, the voltage profile across the extension collapses back to normal — every bus in the extension now lies between 100.0% and 104.0% , against 148% to 210% before compensation.

Transformer loading follows. T1_HVAC drops from 389.2% to 50.9% of its 200 MVA rating and GSU_31 from 96.1% to 83.0%, so no transformer is now overloaded and all critical alerts on the extension are cleared.

What remains is essentially the base system's own weakness. The only critical alerts are Gen_5 and Gen_8 at their reactive limits, both present in the base case and, as noted earlier, not overload conditions. The under voltages at Buses 24, 26, 29 and 30 are unchanged from the base case, so the extension neither helps nor harms the 33 kV periphery. Bus31_400 remains marginally high at 104.0%, indicating the compensation is very slightly under-sized at the sending end, and Glen Lyn 13_1 sits at 103.0% because of its scheduled swing voltage, as before.

Step – 4: HVDC Integration (Uncompensated)

Line-HVAC was taken out of service and replaced by DC_LINK1 between Bus31_400 and BUS1_400, with the SVCs out of service. The load flow converged in 6 iterations.

Table 21: Critical alerts — HVDC uncompensated

Device	Condition	Limit	Operating	%
Gen_1	Over excited	16.219 Mvar	54.320	334.9
Gen_5	Over excited	40.000 Mvar	40.00	100.0
Gen_8	Over excited	50.000 Mvar	50.00	100.0

Table 22: Marginal alerts — HVDC uncompensated

Device	Condition	Limit	Operating	%
Bus31_400	Under voltage	400.000 kV	382.70	95.7
Bus31	Under voltage	132.000 kV	128.29	97.2
Bus 24	Under voltage	33.000 kV	32.320	97.9
Bus 26	Under voltage	33.000 kV	31.80	96.3
Bus 29	Under voltage	33.000 kV	32.07	97.2
Bus 30	Under voltage	33.000 kV	31.68	96.0
Glen Lyn 13_1	Over voltage	132.000 kV	135.96	103.0

Analysis of HVDC Integration (Uncompensated)

The reactive imbalance now runs in the opposite direction to the HVAC case. The DC link generates no charging power, so instead of a surplus the system faces a deficit: the converters draw reactive power from both AC terminals, and total reactive losses are +25.538Mvar against –884.124Mvar in the uncompensated HVAC case.

The result is undervoltage rather than overvoltage, and the depression is confined to the sending end. Bus31_400 falls to 95.7% and Bus31 to 97.2%, both flagged marginals, because the rectifier's reactive demand must be met by G31 alone through two transformers. G31 supplies 40.866Mvar into GSU_31 to hold GEN31_bus at 100.0%, but that support does not reach far enough to hold the 400 kV bus.

At the receiving end the deficit is met by the swing machine, which pushes Gen_1 to 54.320Mvar against a limit of 16.219Mvar, or 334.9%. This is the only critical alert attributable to the corridor. It is far less severe than the HVAC case, where the same machine reached 48 times its capability, but it still exceeds the machine's rating by more than three times and is not an acceptable operating point.

The transformers are all within rating, with T1_HVAC at 53.5%, GSU_31 at 82.9% and T31 at 50.5%. The under voltages at Buses 24, 26, 29 and 30 are again unchanged from the base case.

Compensation is therefore required, but of the opposite type. Capacitive support must be supplied at both converter terminals — larger at the inverter end, where the reactive deficit falls on the swing machine — so that Gen_1 returns within its reactive limit and the sending-end voltages recover.

Step – 5: HVDC Integration with Compensation

SVC31_400 rated 20 MVA at Bus31_400 and SVC_gen_1 rated 45MVA at Glen Lyn 13_1 was placed in service. The load flow converged in 8 iterations.

Table 23: Static VAR compensators — HVDC compensated case

SVC ID	Connected bus	Nominal voltage	Rating
SVC31_400	Bus31_400	400 kV	20 MVA
SVC_gen_1	Glen Lyn 13_1	132 kV	45 MVA

Table 24: HVDC link operating parameters — compensated case

Parameter	Sending end (Bus31_400)	Receiving end (BUS1_400)
Active power, P	100.353 MW	99.089 MW
Reactive power, Q	18.186Mvar	34.478Mvar
Apparent power, S	101.99 MVA	104.92 MVA
Line-to-line voltage	392.8 kV (98.2%)	404.0 kV (101.0%)
AC line current, I	149.9 A	150.0 A
DC voltage / current	655.9 kV / 0.153 kA	
Firing / extinction angle	$\alpha = 5^\circ$	$\gamma = 17^\circ$
Active link loss, P_{loss}	1.2641 MW	

Table 25: Effect of compensation — HVDC case

Quantity	Uncompensated	Compensated
Bus31_400	95.7%	98.2%
Bus31	97.2%	98.5%
GEN31_bus	100.0%	100.0%
BUS1_400	100.6%	101.0%
Glen Lyn 13_1	103.0%	103.0%
Gen_1 output	60.710 MW, 54.320Mvar	58.140 MW, 0.851Mvar
G31 output	117.407 MW, 40.866Mvar	120.461 MW, 21.525Mvar
GSU_31 loading	82.9%	81.6%
T31 loading	50.5%	50.3%
T1_HVAC loading	53.5%	53.5%

Table 26: Alerts — HVDC compensated

Device	Condition	Limit	Operating	%
Gen_5	Over excited (critical)	40.000Mvar	40.000	100.0
Gen_8	Over excited (critical)	50.000Mvar	50.00	100.0
Bus 24	Under voltage	33.000 kV	32.320	97.9
Bus 26	Under voltage	33.000 kV	31.80	96.3
Bus 29	Under voltage	33.000 kV	32.07	97.2
Bus 30	Under voltage	33.000 kV	31.68	96.0
Glen Lyn 13_1	Over voltage	132.000 kV	135.96	103.0

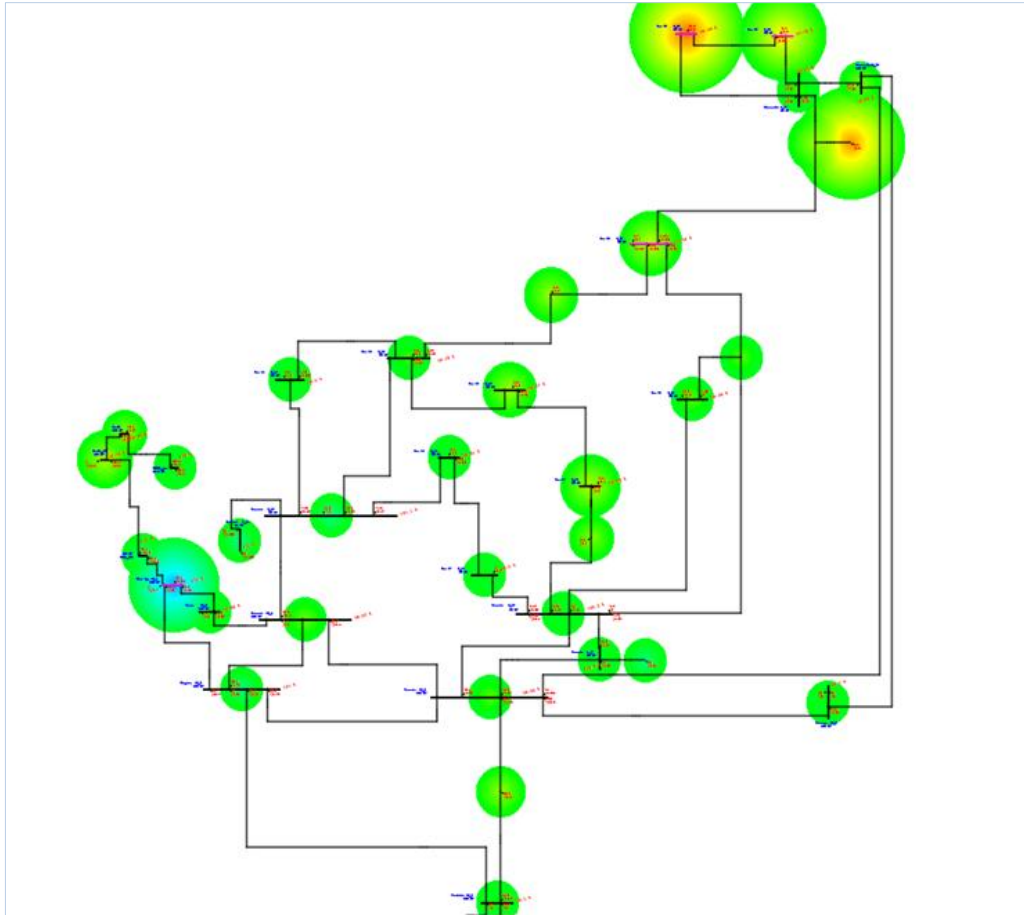


Figure 11: Load flow voltage profile of the HVDC compensated system. Green markers indicate buses within limits; the larger warm-colored markers identify the buses operating at the extremes of the profile

Analysis on Compensated HVDC Integration

Every alert attributable to the corridor is cleared. The under voltages at Bus31_400 and Bus31 no longer appear, and Gen_1 is no longer over excited, leaving only the base system's own Gen_5, Gen_8 and 33 kV undervoltage.

The compensation works by supplying the converters' reactive demand at the point where it arises. SVC31_400 injects 19.3Mvar at the rectifier terminal, which covers the link's 18.186Mvar draw almost exactly. At the inverter end the demand is larger: the link draws 34.478 Mvar at BUS1_400, and with 6.474 Mvar consumed by T1_HVAC itself, 40.952 Mvar must be drawn from Glen Lyn. SVC_gen_1 supplies that locally instead of the swing machine.

The effect on the machines is decisive. Gen_1 falls from 54.320 Mvar, at 334.9% of its 16.219 Mvar limit, to 0.851 Mvar, so the swing machine is now essentially at unity power factor with its full reactive range available. G31 falls from 40.866 Mvar to 21.525 Mvar, retaining margin at the rectifier end.

The voltage profile recovers where it was depressed. Bus31_400 rises from 95.7% to 98.2% and Bus31 from 97.2% to 98.5%, both now clear of the 98% marginal threshold. Across the extension every bus lies between 98.2% and 103.0%, the upper value being Glen Lyn at its scheduled swing voltage. Transformer loading is essentially unchanged, since the SVCs redistribute reactive flow rather than real power.

Voltage Profile and Line loss comparison between Compensated HVAC & HVDC System

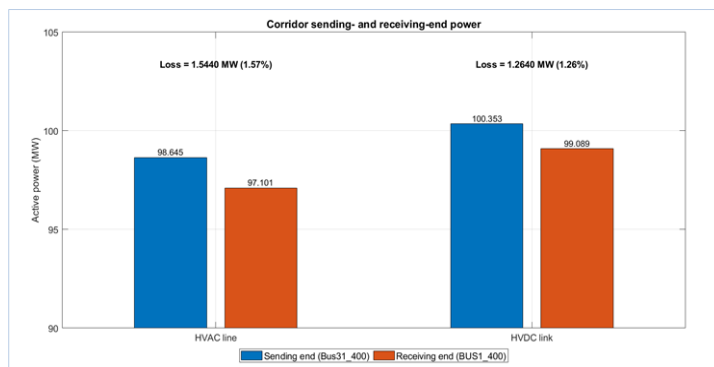


Figure 12: Sending- and receiving-end active power for the compensated HVAC and HVDC corridors, with the resulting transmission loss.

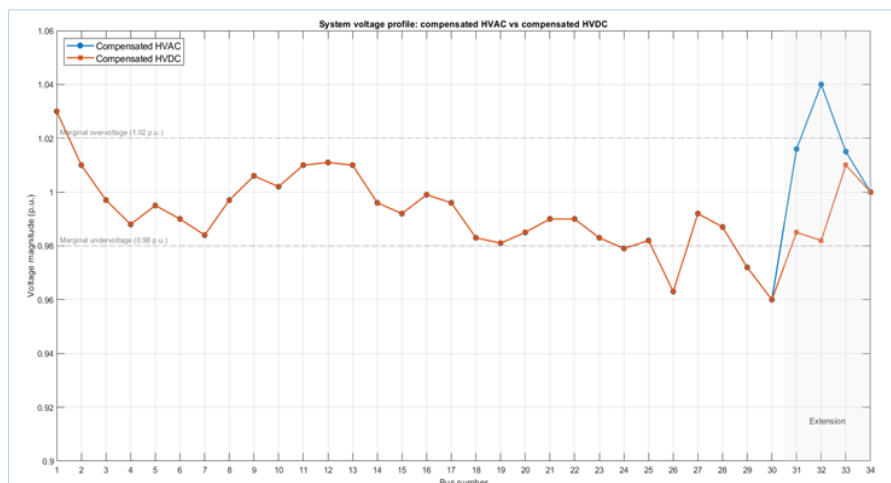


Figure 13: System voltage profile for the compensated HVAC and compensated HVDC cases, with the extension buses (31–34) shaded.

Table 27: Corridor performance — compensated HVAC vs compensated HVDC

Quantity	HVAC line	HVDC link
Sending-end power (Bus31_400)	98.645 MW	100.353 MW
Receiving-end power (BUS1_400)	97.101 MW	99.089 MW
Corridor loss	1.5433 MW	1.2641 MW
Loss as % of sending power	1.565%	1.260%
Sending-end reactive power	−232.001 Mvar	18.186 Mvar
Receiving-end reactive power	−232.540 Mvar	34.478 Mvar
Sending-end apparent power	252.102 MVA	101.99 MVA
Sending-end current	349.9 A	149.9 A
Receiving-end current	358.4 A	150.0 A
Installed compensation	450 MVA (3 reactors)	65 MVA (2 SVCs)
Total system losses	11.072 MW	10.798 MW

The HVDC link transfers 1.708 MW more power than the HVAC line while losing 0.279 MW less, so its corridor loss is 1.260% of the sending-end power against 1.565% for HVAC. Total system losses follow the same ordering, 10.798 MW against 11.072 MW.

The reason is visible in the current rather than the power. Both corridors carry approximately 99 MW, but the HVAC line draws 349.9 A at the sending end against 149.9 A for the HVDC link, because the AC line also carries about 232 Mvar of charging power at each end. Its apparent loading is 252.102 MVA for 98.645 MW of useful transfer, a power factor of 0.391, whereas the DC link's converters draw only 18.186 Mvar at the rectifier and 34.478 Mvar at the inverter. Since loss varies with the square of current, the HVAC corridor pays for reactive current it does not deliver as useful power.

The same imbalance sets the cost of compensation. The HVAC case requires 450 MVA of shunt reactors across three locations to absorb the line's charging power, while the HVDC case needs 65 MVA of static VAR compensation to supply the converters. The HVAC corridor therefore demands nearly seven times the compensation rating for slightly less transfer.

The voltage profiles are identical at buses 1 to 30, so neither corridor affects the original network, and the four under voltages in the 33 kV group persist in both cases. The difference is confined to the extension. The HVAC case leaves Bus31_400 at 1.040 pu., a marginal overvoltage alert, because the residual charging power at the sending end is not fully absorbed. The HVDC case holds the same bus at 0.982 p.u., and the whole extension spans 0.982 to 1.010 pu. against 1.000 to 1.040 pu. for HVAC. The HVDC profile is therefore flatter, better centered on nominal, and free of alerts.

Step – 6: Harmonic Analysis

Compliance Baseline: IEEE 519 Standard The success of the harmonic mitigation was measured strictly against IEEE Std 519 [15], which dictates the maximum allowable voltage distortion at the Point of Common Coupling (PCC).

Table 28: IEEE Std 519 Voltage Distortion Limits at PCC[15]

Bus voltage V (at PCC)	Individual harmonic (%)	THD (%)
V ≤ 1.0 kV	5.0	8.0
1 kV < V ≤ 69 kV	3.0	5.0
69 kV < V ≤ 161 kV	1.5	2.5
161 kV < V	1.0	1.5

Pre-Filter Results and Analysis: Before integrating the harmonic filters, the HVDC converters severely distorted the steady-state voltage waveforms across the network.

Table 29: Pre-Filter Voltage Distortion (THD) at Major Buses

Bus ID	kV	Voltage Distortion Fund. %	Voltage Distortion VTHD %
BUS1_400	400.000	103.00	5.56
Claytor 13_2	132.000	101.00	5.28
Cloverdle13_28	132.000	98.66	2.72
Glen Lyn 13_1	132.000	103.00	5.56

Analysis of Uncompensated System: The pre-filter simulation demonstrated systemic non-compliance. The 400 kV bus (BUS1_400) recorded a THD of 5.56%, eclipsing the 1.5% limit mandated by IEEE 519. Similarly, critical 132 kV buses, including Glen Lyn 13_1 and Claytor 13_2, recorded THDs of 5.56% and 5.28% respectively, far exceeding the 2.5% threshold for that voltage class.

To isolate the source of the distortion, an Individual Harmonic Distortion (IHD) report was generated.

Table 30: Pre-Filter Individual Harmonic Distortion (IHD) at Major Buses

Bus ID	kV	Voltage Distortion VIHD %	Harmonic Order
BUS1_400	400.000	3.40	11.00
Glen Lyn 13_1	132.000	3.40	11.00
BUS1_400	400.000	4.07	13.00
Glen Lyn 13_1	132.000	4.07	13.00

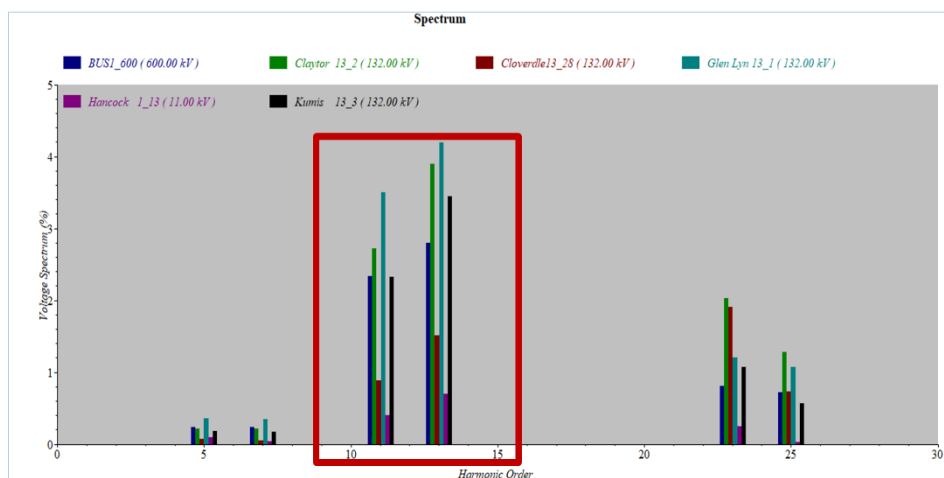


Figure 14: Pre - Filter Harmonic Voltage Spectrum

The THD data confirmed that the 11th and 13th harmonics were the primary source of the fault, registering well above limits at the major terminals. So, we designed 2 single tuned harmonic filter and placed it on Bus 31 and Bus 30(Glen Lyn 13_1).

Post-Filter Results and Analysis: Following the connection of single-tuned filters HF7 and HF3, a subsequent harmonic load flow study was conducted.

Table 31: Post-Filter Voltage Distortion (THD) at Major Buses

Bus ID	kV	Voltage Distortion Fund. %	Voltage Distortion VTHD %
BUS1_400	400.000	103.00	1.76
Claytor 13_2	132.000	101.00	1.61
Cloverdale13_28	132.000	98.66	1.14
Glen Lyn 13_1	132.000	103.00	1.76

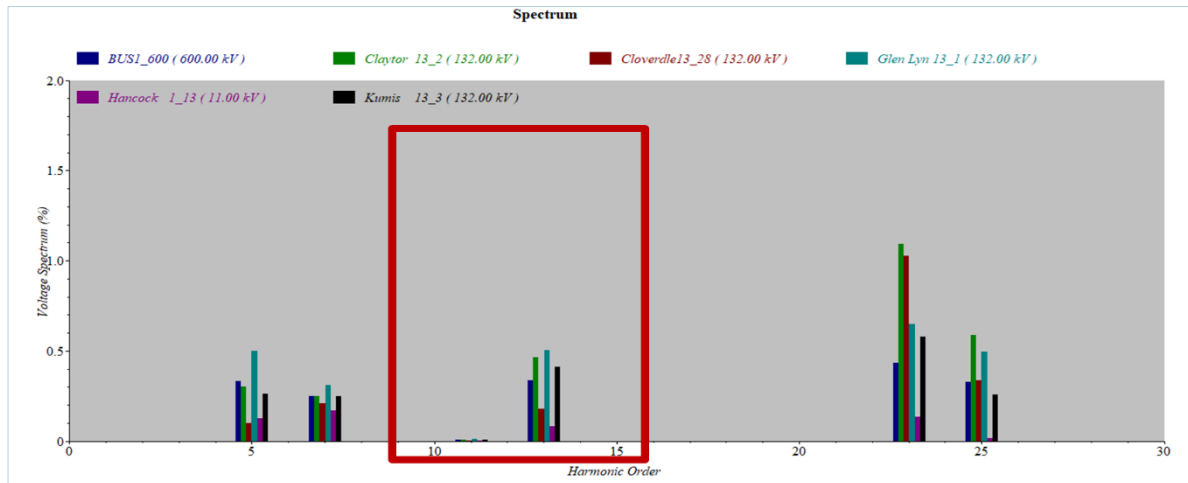


Figure 15: Post - Filter Harmonic Voltage Spectrum

Analysis of Harmonics Compensated System:

The implementation of the passive filters yielded a drastic improvement in power quality. Total Harmonic Distortion plummeted across the system; for example, THD at Glen Lyn 13_1 dropped from 5.56% to 1.76%, bringing it well within the 2.5% IEEE allowance for a 132 kV bus. Spectrum data confirmed that the targeted 11th and 13th order harmonics were virtually eliminated, dropping from over 4.0% to roughly 0.5%.

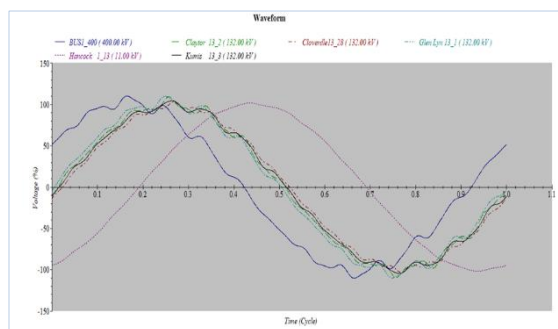


Figure 16: Pre-Filter Voltage Waveforms at Major Buses

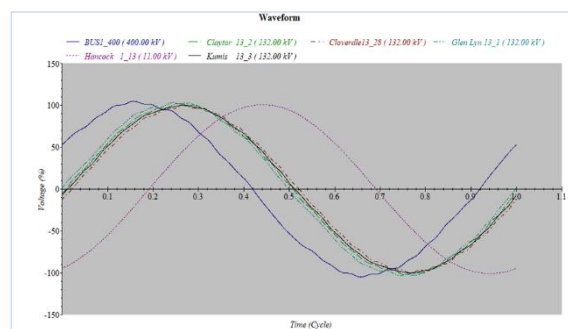


Figure 17: Post-Filter Voltage waveforms at Major Buses

Notably, although the filters were primarily optimized for the 13th order harmonic, their design bandwidth proved sufficient to simultaneously mitigate the 11th order harmonic to safe operational levels, yielding clean, sinusoidal waveforms suitable for safe power delivery.

Step – 7: Contingency Analysis

To check the system's stability in case the HVDC link gets disconnected from system, we performed a contingency test. In real power systems, HVDC AC harmonic filters are connected to the AC busbar using conventional AC circuit breakers (often SF6 gas circuit breakers or high-voltage alternating current circuit breakers) (Csanyi, 2020) ^[8]. These standard AC breakers automatically switch the filter banks on and off to manage changing reactive power levels and harmonic limits on the AC grid. So as soon as HVDC link will get disconnected from the grid, the filters will be switched off.

Table 32: Gen_1 output with the HVDC link in and out of service

Quantity	Link in service	Link out of service	Change
Active power	58.140 MW	157.013 MW	+98.873 MW
Reactive power	0.851 Mvar	7.64 Mvar	+6.789 Mvar
Apparent power	58.146 MVA	157.20 MVA	+99.05 MVA

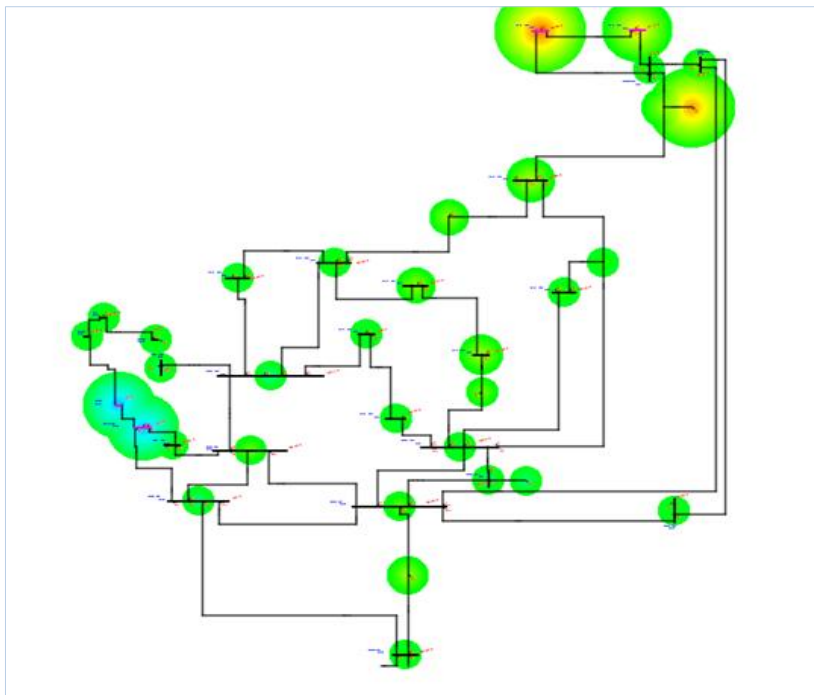


Figure 18: Load flow voltage profile of the contingency test. Green markers indicate buses within limits; the larger warm-colored markers identify the buses operating at the extremes of the profile. The blue markers indicate buses operating in undervoltage conditions.

Table 33: Marginal report — contingency test ((Disconnecting DC link, HF7 & HF3)

Device ID	Type	Condition	Rating/Limit	Operating	% Operating	Phase Type
Bus 26 3_26	Bus	Under Voltage	33kV	31.802	96.4	3-Phase
Bus 29 3_29	Bus	Under Voltage	33kV	32.06	97.2	3-Phase
Bus 30 3_30	Bus	Under Voltage	33kV	31.67	96.0	3-Phase
BUS1_400	Bus	Over Voltage	400kV	412.00	103.0	3-Phase
Glen Lyn 13_1	Bus	Over Voltage	132kV	135.96	103.0	3-Phase

When the HVDC link is out of service, the 98.873 MW it delivered into Glen Lyn 13_1 through T1_HVAC is lost, and the deficit is taken up automatically by the swing machine Gen_1. Its output rises from 58.140 MW to 157.013 MW, an increase of 98.873 MW that matches the lost infeed exactly. At that output the machine is loaded to 60.3% of its 260.698 MVA rating, so it absorbs the entire loss of infeed with substantial reserve remaining. Its reactive output rises only from 0.851 Mvar to 7.64 Mvar, less than half its 16.1 Mvar capability, since the inverter's reactive demand disappears along with the link. The original IEEE 30-bus network therefore continues to operate independently

The generators in the critical report are operating exactly at 100.0% of their rated reactive power (Mvar) capacity, which indicates that their overexcitation limiters have safely clamped the reactive output at its maximum continuous capability rather than actually exceeding thermal limits. Concurrently, the marginal alerts for bus voltages display values ranging from 96.0% to 103.0% of their nominal ratings. Because these values fall comfortably within the standard IEEE/IEC acceptable steady-state voltage tolerance of $\pm 5\%$ (0.95 to 1.05 per unit), the system profile is technically sound.

5.3.4 Interpretation and Conclusions on Data

Why the two technologies fail in opposite ways

The uncompensated results in parts (b) and (d) are mirror images of each other, and the reason lies in how each corridor handles reactive power rather than in any difference in its ability to carry real power.

The HVAC line has a surge impedance of about 287 Ω and a surge impedance loading of roughly 558 MW, yet the study transfers only about 99 MW, some 18% of that value. A long line loaded far below its surge impedance loading generates far more reactive power through its distributed capacitance than it consumes in its series reactance, and the 750 km corridor produces approximately 452Mvar at nominal voltage. With nothing able to absorb it, the terminal voltage rises, which increases the charging power further, and the system runs away to a reactive surplus of –884 Mvar and voltages above 200% of nominal.

The HVDC link inverts this. The DC conductor has no charging current at all, so instead of a surplus the converters impose a demand, drawing reactive power at both terminals in proportion to the power transmitted. The consequence is undervoltage at the sending end rather than overvoltage, and a swing machine driven to 335% of its reactive limit rather than 4800%.

Both configurations therefore require compensation, but of opposite sign and very different magnitude. The HVAC scheme needs 450 MVA of shunt reactors to absorb; the HVDC scheme

needs 65 MVA of static VAR compensation to supply.

HVDC Corridor is more feasible

At comparable transfer the HVDC link loses 1.260% of sending-end power against 1.565% for the HVAC line. The explanation is not in the conductor but in the current each corridor carries. Both move about 99 MW, yet the HVAC line draws 349.9 A against 149.9 A for the DC link, because the AC line simultaneously carries roughly 232Mvar of charging power at each end. Its apparent loading is 252.102 MVA for 98.645 MW of useful transfer, a power factor of 0.391. Since resistive loss varies with the square of current, the AC corridor pays for reactive current that delivers no power to the receiving end.

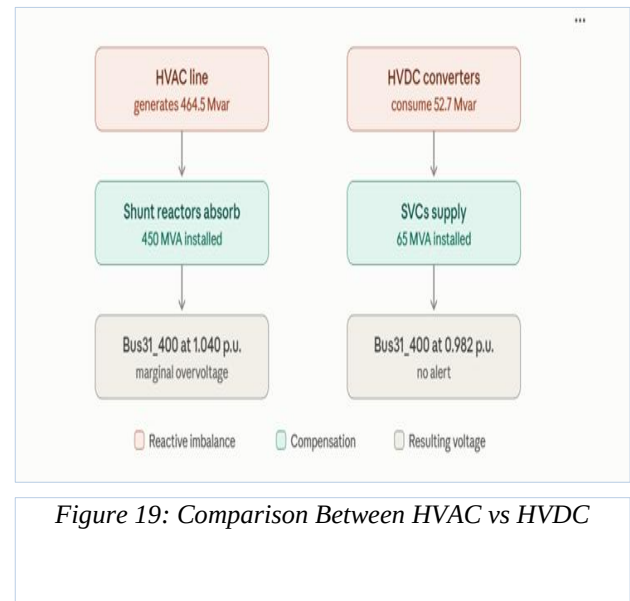


Figure 19: Comparison Between HVAC vs HVDC

Quality of the resulting Voltage profile

After compensation both schemes are workable, but they are not equally good. The HVDC case holds the whole extension between 0.982 and 1.010 p.u. and raises no alert on any bus of the corridor. The HVAC case leaves Bus31_400 at 1.040 p.u., a marginal overvoltage, because the residual charging power at the sending end is not fully absorbed by shunt31. The HVDC profile is flatter and better centered on nominal, and it achieves this with less than one seventh of the installed compensation.

The Corridor cannot change certain bus profile and generation

The voltage profiles of buses 1 to 30 are identical in the two compensated cases and unchanged from the base system. The four marginals under voltages at Buses 24, 26, 29 and 30, and the reactive-limited operation of Gen_5 and Gen_8, persist in every configuration studied. This is an important negative result: reactive support does not travel well through a network, and adding 120 MW of generation at a remote 400 kV terminal does nothing for a weak 33 kV radial group fed from Cloverdle. Correcting that deficiency requires local compensation in the 33 kV group, not a stronger corridor.

Conclusions

For this corridor and this loading, HVDC is the better choice on every measure examined. It transfers more power, loses less of it, produces a flatter voltage profile, raises no voltage or loading alerts, and requires substantially less reactive compensation to reach an acceptable operating point.

The conclusion is specific to the operating condition studied, however, and should not be generalized without qualification. The comparison is made at about 18% of the AC line's surge impedance

loading, which is the regime in which an AC line performs worst, since its charging power dominates and must be absorbed. Loaded closer to its surge impedance loading the same line would be approximately self-compensating, its charging power balanced by the reactive consumption of its own series reactance, and much of the disadvantage identified here would disappear. The result therefore demonstrates that HVDC is superior for transmitting a modest power over a long distance, not that it is superior at all loadings.

Three matters affect how strongly these conclusions can be stated. G31 is dispatched at a fixed 120.000 MW in the HVAC cases but operates as a swing machine in the HVDC cases, so the two configurations are not transferring exactly equal power; the difference of 1.7 MW is too small to reverse the result but should be eliminated. SVC_gen_1 appears to operate at approximately 47.7 Mvar against a 45 MVA rating, so the HVDC compensation may be slightly undersized. And the DC link is rated 112.2 MW while G31 is rated 149.714 MW, so the link as designed could not export the machine's full output.

5.4 Limitations of Tools (PO(e))

We began this project in PSS/E, using it for the base IEEE 30-bus model and the first HVAC load-flow runs. PSS/E is built for power flow, short-circuit and dynamic studies, but it has no built-in module for designing or sizing harmonic filters — Siemens keeps that capability in a separate product, PSS SINCAL, not PSS/E [16]. Harmonic mitigation is central to this project, since the HVDC converters inject 11th- and 13th-order harmonics that our filters had to be designed and checked against, so we moved the whole study to ETAP partway through.

ETAP handles load flow, short-circuit, harmonic load flow and IEEE 1531-based filter sizing in one place [15], so the HF7 and HF3 filters in Section 4.7 could be designed and checked without switching tools again. The change cost time, but every result in this report comes from ETAP.

A few other tool limitations are worth flagging:

- **Steady-state only:** Our load-flow, harmonic and short-circuit studies are frequency-domain snapshots. They cannot show what happens in the first few milliseconds after a switching event, energizing the HVDC link or a filter, or clearing a fault, which would need a time-domain (EMT) tool such as PSCAD.
- **No automatic compensation sizing:** ETAP flags marginal and critical violations but does not suggest a fix for them. Every shunt reactor, SVC and filter rating in this report was reached by trial and check, which is why some margins are thin, for example the SVC at Glen Lyn 13_1 operates close to its rated capacity rather than with headroom (Section 4.3.4).
- **Balanced, positive-sequence only:** The model inherits the IEEE 30-bus system's balanced three-phase representation, so unbalanced faults and single-phase effects were not modelled.

5.5 Impact Assessment (PO(f))

Our simulation showed that HVDC link can significantly reduce line loss compared to HVAC in case of distant transmission. A plant sited 750 km from load can still be delivered to the grid within acceptable voltage and loss limits, provided the corridor is properly compensated and, for the HVDC

option, its harmonics are filtered. That is the actual result worth assessing, and it maps onto a real problem for Bangladesh, where new generation, gas, offshore wind, remote solar, is increasingly built away from the existing Dhaka-Chittagong load corridor.

Two other ideas came up while researching this section: coupling solar generation straight into an HVDC link without inverting to AC first, and feeding large DC loads such as data centres directly off a converter station. Both are being explored in the literature ^{[11], [12]} at an initial stage, which could be expanded in future work.

5.5.1 Assessment of Societal and Cultural Issues

Connecting a stranded plant like the one at Bus 31 is the same problem Bangladesh faces as it adds generation away from its existing load centers; this project shows that both the HVAC and HVDC options can do it over 750 km, with HVDC holding a small edge

5.5.2 Assessment of Health and Safety Issues

Before filtering, voltage distortion reached 5.56% at some buses above the 1.5-2.5% IEEE 519 allows for this voltage class, enough to cause nuisance relay trips and extra transformer heating for anyone fed from those buses. The HF7 and HF3 filters bring every bus back under its limit, which is the actual safety outcome of the harmonic study. Electric and magnetic field exposure along the corridor itself is covered in Section 4.6.

5.5.3 Assessment of Legal Issues

The filters were sized against IEEE Std 519 ^[15], not a rule of thumb, so the design has a real compliance basis. A corridor built this way in Bangladesh would also need to clear the Bangladesh Energy Regulatory Commission's Electricity Grid Code, which governs interconnection and right-of-way approval ^[14].

5.6 Sustainability Evaluation (PO(g))

Replacing the 750 km HVAC corridor with the bipolar HVDC link modelled in Section 4.1.4 has two direct environmental effects: less corona discharge, and less conductor material for the same power transfer.

Corona effect

- Corona begins once the electric field at the conductor surface passes the air's critical disruptive gradient, of the order of 30 kV for the conductor geometry used here ^[10].
- The discharge ionizes the surrounding air, giving off light, heat, an audible hiss, and a small amount of ozone (O₃) as it splits and recombines oxygen molecules ^[10].
- An AC conductor crosses that threshold on every half-cycle and discharges from all three phases, roughly a hundred times a second; a DC conductor holds a constant polarity and discharges only from its positive pole.
- Measured HVDC lines show radio interference about 6-8 dB lower than an equivalent HVAC line, and, unlike AC, DC corona performance improves rather than worsens in rain ^[9].

- Net effect: moving this corridor to HVDC should mean less sustained corona-driven ozone and noise along the right-of-way than the HVAC case examined earlier in this report.



Figure 20: Corona discharge of a 500kV Overhead Transmission Line.[10]

Less conductor material

- A bipolar DC link needs two conductors against three phase conductors for the equivalent AC line, a one-third reduction that also means lighter towers and a narrower right-of-way ^[9].
- This project's own results show the wider equipment saving that follows: the 750 km HVAC line needed 450 MVA of shunt reactors at three separate locations to stay within limits, while the HVDC link needed only 65 MVA of SVC at its two converter terminals, roughly a seventh of the equipment for a comparable power transfer (Section 4.3.4).
- Less conductor material and fewer, smaller compensation banks per kilometer mean a smaller embodied-material and land footprint over the life of the link — weighed, as Section 4.3.4 already discusses, against the converter stations' own equipment and losses.

5.7 Ethical Issues (PO(h))

All ETAP simulation models were developed independently by the authors using ETAP's built-in IEEE 30-bus system library. Online writing assistance was used strictly to refine and polish the documentation's tone, while all simulation results, technical analyses, and draft contents were created and verified by the team. We later verified the work with iThenticate to check plagiarism and percentage was less than 20%.

6 Reflection on Individual and Team work (PO(i))

6.1 Individual Contribution of Each Member

The project was carried out by a six-member team (Student IDs 2206013-2206018), organized into four sequential work packages: theoretical study, simulation build-up, mitigation and compensation design, and project finalization. Tasks were distributed as evenly as possible across the team so that each member led at least one technical work package end to end, while the discussion/planning session, the overall-outcome synthesis and the report and presentation preparation were carried out jointly by all six.

Student ID	Primary Responsibility	Key Tasks Completed
2206013	Load-flow modelling and simulation	Built and validated the IEEE 30-bus and HVAC load-flow cases in ETAP; carried out the compensated-HVAC overvoltage correction; supported the harmonic load-flow runs and the environmental-impact .
2206014	Report integration	Report writing,organizing and Base case implementation.
2206015	HVDC modelling and compensation	Led the HVDC theoretical study and the IEEE 30-bus build-up; ran the HVDC load-flow case and its shunt/SVC compensation; contributed to the cost analysis.
2206016	Harmonic analysis, filter design and cost analysis	Led the harmonic analysis in ETAP and the design of the two single-tuned filters (HF7, HF3); supported the harmonic simulation runs and the cost analysis.
2206017	Harmonic analysis and environmental impact	Co-led the harmonic filter design and simulation; led the environmental-impact assessment and supported the IEEE 30-bus build-up and the cost analysis.
2206018	Contingency analysis	Led the contingency study in ETAP from the theoretical stage through simulation and the SVC-based mitigation design used in Section 4.3.3(g).

All six members took part in the discussion and simulation-planning session, the overall-outcome synthesis, and the preparation of the report and presentation.

6.2 Mode of Team-Work

The team combined in-person and remote collaboration. During the mid-semester break, two full-day offline sessions were held in the students' discussion space in the ECE Building, running from morning to evening, during which the group worked jointly on the ETAP model, the harmonic filter design and the contingency study. On other days, the team met online through Google Meet, and day-to-day coordination, file-sharing and quick queries were handled through a dedicated Telegram group, which kept the whole team synchronized between the scheduled meetings.

6.3 Diversity Statement of Team

All six members are undergraduate students from the same EEE cohort, and the team's diversity lies mainly in complementary technical interests: load-flow and compensation modelling, harmonic analysis and filter design, contingency and protection studies, and cost analysis and technical writing. Work was assigned along these interests so that each member could contribute from an area of relative strength, while every major design decision, the choice of compensation scheme, the filter tuning order, and the contingency mitigation, was agreed jointly in the team meetings described in Section 4 and 5.

6.4 Log Book of Project Implementation

Date	Milestone achieved	Individual Role	Team Role	Comments
18.07.26 - 19.07.26	Theoretical study and simulation plan completed for all four work-streams	Each member completed the theoretical review for their assigned stream	Joint discussion and simulation planning (all members)	Kick-off offline meeting, ECE Building discussion space
19.07.26 - 26.07.26	IEEE 30-bus model, HVAC / HVDC load flow and harmonic load flow simulations completed in ETAP	2206013, 2206015: load flow; 2206016, 2206017: harmonic; 2206018: contingency set-up	Simulation build-up	Base and extended ETAP model cases converged
26.07.26 - 09.08.26	Overvoltage compensation, harmonic filter design and contingency SVC mitigation completed	2206013, 2206015: compensation; 2206016, 2206017: filters; 2206018: contingency mitigation	Mitigation and compensation design	Second offline meeting, ECE Building discussion space
09.08.26 - 20.08.26	Environmental-impact and cost analysis, overall-outcome synthesis, report and presentation completed	2206017, 2206013: impact; 2206016, 2206018, 2206015: cost; all members: outcome and report	Project finalization	Coordinated remotely via Google Meet and Telegram

7 Communication to External Stakeholders (PO(j))

7.1 Executive Summary

This report evaluates the integration of a remote 150 MW power plant into the IEEE 30-bus system over a 750 km corridor using ETAP software. It compares properly compensated High Voltage Alternating Current (HVAC) and High Voltage Direct Current (HVDC) transmission technologies. The study finds that HVDC is the superior choice for this distance, demonstrating lower

transmission losses (1.264 MW versus 1.543 MW) and requiring significantly less reactive compensation (65 MVA versus 450 MVA). While HVDC converters introduce harmonic distortion, implementing single-tuned passive filters effectively restores power quality to comply with IEEE Standard 519.

7.2 User Manual

To demonstrate the complete project one should obey the following user manual. You can also follow the following YouTube video for better understanding.

Step 1: Base Case Setup

- Open the ETAP 16.0.0 software and construct the standard IEEE 30-bus test system.
- Run a Newton-Raphson load flow study to verify that the base system converges reliably and operates without any critical alarms.

Step 2: System Extension (Adding Remote Generation)

- Create four new buses to expand the network: GEN31_bus (13.8 kV), Bus31 (132 kV), Bus31_400 (400 kV), and BUS1_400 (400 kV).
- Connect a new 150 MW generating unit (G31) at the GEN31_bus terminal.
- Add a local 20 MW load at Bus31.
- Install transformer GSU_31 (150 MVA) and transformer T31 (200 MVA) to step the voltage up to the 400 kV transmission level at Bus31_400.
- At the receiving end, connect BUS1_400 to the existing 132 kV Glen Lyn bus using the T1_HVAC transformer (200 MVA).

Step 3: HVAC Transmission and Compensation

- Connect a 750 km, 400 kV HVAC line between Bus31_400 and BUS1_400.
- To absorb the high charging reactive power (Ferranti effect), install three fixed shunt reactors: 170 MVA at Bus31_400, 250 MVA at BUS1_400, and 30 MVA at Glen Lyn 13_1.
- Run a load flow study to confirm that all bus voltages stabilize within the safe limits of 0.95 to 1.05 per unit.

Step 4: HVDC Transmission and Compensation

- Take the HVAC line out of service.
- Replace the connection with a 660 kV bipolar HVDC link (DC_LINK1) between Bus31_400 and BUS1_400.
- Provide local reactive support for the converter stations by installing Static VAR Compensators (SVCs): one rated at 20 MVA at Bus31_400 and another rated at 45 MVA at Glen Lyn 13_1.
- Run the load flow study to record the system voltage profile and transmission losses.

Step 5: Harmonic Filter Implementation

- Run a harmonic load flow study on the HVDC configuration to identify the 11th and 13th order harmonic distortions.
- Design and connect two single-tuned passive filters configured as series R-L-C branches (HF7 at Bus 31 and HF3 at Glen Lyn 13_1) with a quality factor of 20.
- Re-run the harmonic load flow study to verify that the Total Harmonic Distortion (THD) is reduced below the IEEE Standard 519 limits (1.5% for buses above 161 kV and 2.5% for

69-161 kV buses).

Step 6: Contingency Analysis

- Configure a contingency test in ETAP simulating the sudden disconnection of the HVDC link from the grid.
- Ensure that the conventional AC circuit breakers automatically switch off the associated harmonic filters during the disconnect event.
- Run the simulation to confirm that the network survives the outage and no critical bus voltages or equipment overloads occur.

7.3 Github Link

Github Link of the Complete project:

[nurealam013/Feasibility-of-HVDC-Transmission: Feasibility of HVDC Transmission for Integrating a Remote 150 MW Generation Source into the IEEE 30-Bus System.](https://github.com/nurealam013/Feasibility-of-HVDC-Transmission: Feasibility of HVDC Transmission for Integrating a Remote 150 MW Generation Source into the IEEE 30-Bus System.)

7.4 YouTube Link

Part – 01: HVAC and HVDC implementation in Standard IEEE 30 – Bus

<https://youtu.be/eayd7xh9XXc?si=1qyCEeeR9yPc2rNz>

Part – 02: Harmonic Analysis <https://youtu.be/TQLWsrzGJvc?si=4sLmij6hc9WSQQN>

Part – 03: Contingency Analysis <https://youtu.be/vpEwVKrKKDE?si=pYdSDwpA-Pvz4IKH>

8 Project Management and Cost Analysis (PO(k))

8.1 Bill of Materials

As the project is implemented in ETAP, there is no bill of the power system components.

8.2 Calculation of Per Unit Cost of Prototype

The prototype of this project is the ETAP model itself, not a piece of hardware. Its cost is therefore the cost of the software, the computers and the engineering time that went into building and studying the model.

Table 36: Cost of building and studying the simulation model (the prototype)

No.	Item	Basis	Cost to the project (Tk)
1	Computers	6 laptops @ Tk 80,000, 5-year life, used for 5 weeks(depreciation)	9,231
2	Electricity and internet	5 weeks	500

3	Printing, binding and stationery	Report and presentation	500
	Total notional cost of the prototype		10231

8.3 Calculation of Per Unit Cost of Mass-Produced Unit

We are calculating the cost of the components to add in the grid to stabilize the system. The cost is written below:

Bank	Installed rating (kvar)	Total cost (Tk)crore
Filter 1 (HF7)	19,503	4.86,1
Filter 2 (HF3)	63,686	7.77,4
SVC31_400	20,000	22.5
SVC_gen_1	45000	10
Both filters together	83,189	22.135

8.4 Timeline of Project Implementation

Task no	Task	Start Date	End Date	Week 1	Week 2	Week 3	Week 4	Week5
1	Theoretical Study	12.07.26	19.07.26					
2	Simulation	19.07.26	26.07.26					
3	Mitigation	26.07.26	09.08.26					
4	Project Finalising	09.08.26	20.08.26					

9 Future Work (PO(I))

This project answered the main question, but several parts of it can be taken further.

- **Using a VSC-HVDC link instead of an LCC link.** A voltage source converter can control real and reactive power independently, so it would not need the two SVCs that the present design needs, and its higher pulse number would produce much smaller harmonics. Comparing an LCC design and a VSC design on the same network would be a direct extension of this work.
- **Implementing transient and EMT studies.** Only steady state and harmonic load flow studies were done here. A transient stability study would show how the system behaves in the first few seconds after a fault, and an electromagnetic transient study would show commutation failure, DC line faults and the real switching behaviour of the converters.
- **Finding the break-even distance for this network.** The corridor was fixed at 750 km. Repeating the whole study at 300, 500, 750, 1000 and 1500 km would show exactly where HVDC starts to win for this system, and a full life-cycle cost comparison including the converter stations would turn that into a real economic answer.
- **Optimising the compensation instead of tuning it by hand.** The reactor and SVC sizes were found by looking at the ETAP alert reports and trying values. An optimisation routine could find the smallest total MVAR and the best locations automatically.
- **Trying active and hybrid filters.** Passive single-tuned filters are cheap but they are fixed. They can resonate with the network if the system impedance changes and they always inject reactive power. An active or hybrid filter would adapt to changing conditions and could be compared on cost, loss and performance.
- **Designing the protection scheme.** The DC line and the converters need their own protection. Travelling wave and DC resistance-based protection schemes for a bipolar line could be designed and tested on this model.
- **Making the remote source a renewable plant.** The 150 MW source was modelled as a conventional generator. Replacing it with a wind farm or a solar park would add variable output and would test how well the link and the filters cope when the power flow changes all day.
- **Extending the link to more than two terminals.** A multi-terminal DC network with a third tap point would be more useful and much more difficult to control, and is a natural next step.
- **Performing a reliability and availability study.** The contingency study looked at one event. A proper reliability study would put failure rates and repair times on the link, the converters and the filters and calculate the expected energy not supplied.

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